

AMERICAN BEE JOURNAL.

EDITED AND PUBLISHED BY SAMUEL WAGNER, WASHINGTON, D. C.

AT TWO DOLLARS PER ANNUM, PAYABLE IN ADVANCE.

VOL. VI.

OCTOBER, 1870.

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AMERICAN BEE JOURNAL, VOL. VI, NO. 4, OCTOBER 1870 ***

AMERICAN BEE JOURNAL.

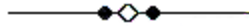
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[Translated for the American Bee Journal.]

Origin of Honey Dew.

In No. 11 of the *Bienenzeitung* for 1870, the Baron of Berlepsch urges bee-keepers to make diligent observations, to ascertain the origin of honey dew. I have for many years given special attention to the subject, as it is one of great interest, not only to bee-keepers, but also to pomologists. My observations fully corroborate the remark of the Baron, that honey dew occurs, in most cases, independently as a vegetable excretion, and only occasionally as the product of aphides. On last Sunday, June 19th, I had an opportunity to assure myself definitely of the correctness of this position. On that day, as early as seven o'clock in the morning, I received a visit from Mr. Heuser, of Westom, one of the intelligent apiarians who compose the Ahrweiler Association for Bee-culture. While we sat conversing about bees, a lad came to inform us that he had, the evening before, seen a fine swarm clustered on a large pear tree. We naturally hastened to the spot, but found that the swarm had already decamped. A loud humming among the branches, however, led us to suppose there might be a hollow limb somewhere, into which the bees had retreated, and friend Heuser was induced to climb up in search of it. He found none, but observed a multitude of bees busily engaged licking up the honey dew with which the leaves of the tree were covered—being evidently an exudation, for on the most careful examination we could not find a single aphid, though on the morning of the next day thousands of aphides were observable there.

It remains for me to mention the state of the weather at the time, for according to my observations this chiefly conditions the production of honey dew. On Saturday, June 18th, the weather was oppressively hot. Towards evening the wind began to blow from the northwest; and the night was cool, though without dew on the grass. This necessarily checked the circulation of sap, which I regard as the primary cause of honey dew, for I may state explicitly that I never saw any, except when hot days were followed by a sudden and great reduction of temperature. The same observation was made, many years ago, by an aged bee-keeper in Niederheckenbach, who, whenever he notices in summer a sudden change

of weather, at night, from great heat to cold, will rise at three or four o'clock in the morning and close the entrances of his hives; as he is firmly persuaded that the honey dew certain to come, will be injurious to his bees. I must confess that honey dew has not always proved beneficial to our bees. In some cases they seemed to be sickened by it, and to remain so for nearly a week, as indicated by their inability to fly. This was more especially the case at an apiary which I had in an oak forest, where bark was largely stripped and dried for tanners' use. I am unable to account for the occurrence, and must leave chemists to determine whether the consumption of *tannin* had aught to do with it. Whenever honey dew occurs in my neighborhood again I will strip leaves from various trees affected by it, and send them for examination to Dr. Keermrodt, of Bonn, the chemist of the Agricultural Experimental Union of the Rhine province.

The views of Prof. Hallier, that the honey dew produced by aphides is of great practical account in bee-culture, I am not prepared to endorse. During the summer of 1869 I was a student in the Pomological Institute at Reutlingen, and very seldom saw a bee on any twig covered with aphides, yet we were there sorely annoyed by those parasites. Even now, I am compelled to use soapsuds, &c., to rid my plants of these unwelcome visitors, yet I have never seen a bee among them.

Your readers will probably be interested in learning the views of two of the most eminent pomologists, regarding the origin of honey dew.

Court-gardener Jager, of Eisenach, writes as follows to Regel's Garden-Flora:—"According to my observations, honey dew is much more frequently exuded from the leaves of plants than produced by aphides. I regard honey dew, in many cases, as *a segregation of the saccharine portion of the juices of plants, which these are then no longer able to excrete out of their organism by means of the blossoms*. I was led to adopt this view by repeatedly observing that linden trees so kept under by pruning that they never blossom, excrete such a superabundance of honey dew that such as is not gathered by insects, drips from the leaves to the ground, and is often collected on boards and bottled. Linden trees which are allowed to blossom, do indeed likewise produce honey dew; but I have never seen it on

trees that bloomed profusely, and as I live in the midst of lindens, I have the best opportunities for observation.”

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Next, my own respected teacher, Dr. Lucas, of Reutlingen, remarks, in a note on the foregoing passage—

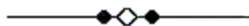
“This observation of our esteemed friend Jager certainly deserves attention. Whether he is entirely right or not, is to me not altogether clear. I have seen honey dew indiscriminately on young trees and on old of various kinds; but always only after we had several successive hot and dry days, followed by dewless nights. It is very probable that then the juices of plants become more concentrated, and thus more highly charged with saccharine, in so much that drops of liquid sweet may exude through the pores of the leaves, and that then the aphides will quickly resort to the tables thus ready decked for them, and multiply with almost incredible rapidity, is a natural phenomenon observable in the case of other insects also. But that the aphides are the originators of the honey dew, as many foresters and others maintain, can certainly not be accepted as correct and true.”

Allow me, in conclusion, to request bee-keepers and pomologists to watch for the appearance of honey dew on the occurrence of such weather and temperature as above indicated, and to communicate the result of their observations.

A. ARNOLD,

*Travelling Lecturer of the Agricultural Union,
Province of the Rhine.*

Löhndorf, June 22, 1870.



[For the American Bee Journal.]

Profitable Bee-keeping.—Letter from England.

The following account shows the very great advantage in keeping bees on the humane and improved system, over the old and barbarous practice of the brimstone match, so clearly, that I send it for your readers to go and do likewise.

In the autumn of 1865, I was at the seaside on the Lancashire coast, and found bees kept in that neighborhood in the most primitive and bad way I ever met with in any country. It was the system there to put the swarm in a large brown wicker basket, and at night to plaster a thin coating of cowdung over the outside, and leave it in this way all summer. I have frequently seen the bees coming out of holes all over the hive, from top to bottom, not being able to fill up all the nicks with propolis, and giving it up as a bad job; and if it was not a good district for honey, they would give up the ghost altogether.

When the bees give over working, the owner plasters the hive with mortar, for the winter. The entrance is made three or four inches high from the cold slate or flag on which they place the basket. When they take the honey, they suffocate the bees with brimstone. Wasps often destroy the stock.

In my perambulations I called upon a person who had kept bees for a number of years in the old way; but they had all died off except one stock. After talking with him for some time on the humane and profitable management of his bees, and showing him the great loss that he sustained by murdering his poor bees, to say nothing of the ingratitude or sin in killing them after they had been laboring for him early and late all the summer, and proved to him the very great advantage the modern bar-frame (thanks to the Rev. L. L. Langstroth, the inventor) from which the honey could be taken without killing a bee, and swarms made or prevented, as we liked. I showed him that in fact, with these hives, he had the full control over his bees, and could make them do almost anything he liked.

He asked me to get the man that makes my improved bar-frame hives, to send him some; and I afterwards sent him information he wrote for in several letters.

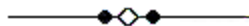
When I called on him last October, I found twenty stocks of bees in his garden, all very strong, with plenty of honey to last them over the winter; and he had sold nearly *three hundred weight of honey*, all of which he had taken that year, *without killing a bee*. He has now got his stock up to the number he intends to keep, so this year he will work for honey; and if it is a favorable season, his bees will collect for him an immense store and make him a nice addition to his income.

The same year that I called upon him, I called upon his neighbor, a person much better off than the other, and he then had three stocks of bees. I advised him to adopt the more profitable and humane system of management; but he did not; and when I called on him again last October, I found three weak stocks of bees in his garden, and he said he had taken *no honey* that year and got very little the year before. I turned his hives over and found an accumulation of wet filth and dirt, nearly an inch thick on the slate floors on which his hives were placed, and the bottoms of the combs all mouldy.

I told him if he had done as well as his neighbor, he should now have sixty stocks of bees in his garden and have taken more than a thousand weight of honey that year. He is now, with others in that district going to adopt the humane system of management, and I hope bee-murder has forever disappeared in that locality, as I always find, when they see the loss to their own pockets, it is the most convincing argument that can be used.

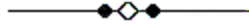
WILLIAM CARR.

Newton Heath, near Manchester, England.



Bees sometimes abandon their hives very early in the spring or late in the summer or fall. They exhibit all the appearance of natural swarming; but they leave not because the population is crowded, but because it is either so

small, or the hive so destitute of supplies that they are discouraged or driven to desperation. I once knew a colony to leave a hive under such circumstances, on a spring-like day in December! They seem to have a presentiment that they must perish if they stay, and instead of awaiting the sure approach of famine, they sally out to see if something cannot be done to better their condition.—*Langstroth*.



[From the Western Farmer.]

About Patents.

A student in the Michigan Agricultural College has invented a gate latch, for which he has received \$10,000.

We find the above item in our exchanges. Assuming it to be true, we commend the good sense of the student. If the usual results follow, the purchaser will either lose money by the operation, or will speedily sell “rights” to parties who will lose money. We have no wish to discourage inventors, for they certainly are entitled to full reward for any improvements or discoveries they give the world. But we think it is clearly true that the great mass of inventors—especially those whose inventions relate to “little things,” or articles in common use—place too high an estimate on the value of their patent right, often holding it, waiting for better offers from manufacturers or purchasers of “territory,” until some one patents a better device for the same purpose, when the first becomes useless or nearly so.

There are certain inventions of very great value, because they supply a want universally felt. But even in such cases it is rare that the original inventor secures so high a degree of excellence that some one else cannot improve on his device. He may, however, succeed in patenting something which subsequent inventors will have to use, and for which privilege they must pay him. To illustrate: the plow is of almost universal use, yet there are objections to the best plow that has been or will be constructed. Suppose some one should invent an implement that would obviate all these objections, and do the work of preparing the soil for seeds better than any plow can, and do this work quickly and cheaply. Such an invention would be of almost incalculable value, and the inventor might well expect to become very wealthy. Yet it would be strange if some one did not improve on this invention, and thus divide the profits—perhaps take the larger share. Hundreds of men have suggested improvements of more or less value in reapers, after the main principle had been given to the public.

In case of such an invention as a gate latch, it must be remembered that there are already very good ones in existence, and probably a still better one may soon be invented; and so we say that, in all ordinary cases, it is better to sell the patent if any such price as \$10,000 is offered for it. However useful such an invention may really be, the inventor as well as the intending purchaser of a “right” should carefully avoid forming extravagant opinions as to “the money there is in it.” The farmer or other business man who gives up his regular business to engage in the sale of patents, in the great majority of cases, does a very foolish thing.

We write this, because we have noticed in many cases the high anticipations of inventors or of purchasers of “territory” for some patent, and the disappointment and loss that followed. If any of our readers have invented anything they are convinced is of value, we say patent it by all means; but do not think of leaving your farm or other business to engage in its sale, or dream of sudden wealth to come from it.



[For the American Bee Journal.]

Hurrah for 1870, and the Honey-slinger.

The best honey season on record, and the most useful invention! Long live our German friend, who gave it to us without a patent!

The battle is past, and we can look back and see if the generalship has been, like that of the Prussians, well managed—or, like that of the French, left to manage itself.

I had two stocks last spring, and the empty combs from two hives that died about the first of March. The first swarm was hived on the 18th of June, and the honey-gathering on bass-wood closed July 26th—so that none of the young bees in new hives were then old enough to gather honey.

I have taken one hundred and eighty-seven (187) pounds with the machine, and on the 26th of July had from five hives, 228 lbs., or forty-five pounds each. They had gained forty pounds each, in thirteen days, on bass-wood blossoms. The best stock gained, 52 lbs. 8 oz. A queenless stock gained 33 lbs. 10 oz. The best day's work, 7 lbs., Aug. 16. The best day's work in June was Saturday and Sunday, the 25th and 26th—a gain of 21 lbs. 6 oz. on red raspberry blossoms, or 10 lbs. 11 oz. per day. I see that NOVICE reports 43 lbs. in three days, 25th, 26th, and 27th of June. As he reports bass-wood at its best July 6th, the flowers must be ten or twelve days earlier than at this place. So his best yield of honey, on the same days as mine, at 600 miles distance, was perhaps on account of the weather, or some electrical state of the atmosphere.

In June I took from my stocks what honey they had above twenty pounds each. While bass-wood was in blossom, I tried to take what they had above forty pounds each. The honey-emptier appeared to take away all disposition to raise a lot of drones in July. When I depended on box honey, the hive was crowded with honey before the bees would work in boxes.

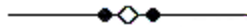
As it took two pounds per month in winter to support a colony of bees, at this rate the twelve ounces of honey required to rear a thousand drones

would keep a thousand workers four and a half months. I believe drones usually live about two months. So when NOVICE shaves off the heads of drone brood sealed over, he has already lost two-thirds of what it would cost to let them live; and the presence of drones might perhaps prevent the raising of more drone brood.

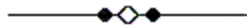
I would like to have NOVICE answer one question through the BEE JOURNAL, and that is—Do light queens make better honey-gathering stocks than dark queens from the same parents?

HENRY D. MINER.

Washington Harbor, Wis.



A charlatan is an impostor who lives by the folly of those who are imposed upon.



[For the American Bee Journal.]

Comments on Querist No. 7.

On page 83, Vol. V., of your most valuable journal, Querist seems to be at variance with our position in an article on page 55, of the same volume, where we assumed, as we yet maintain, that “the first and highest law of nature in insects is self-preservation in caring for offspring, &c. The honey bee seems to be endowed with this instinct for the purpose of preserving the brood in the hive.” Querist asks—“Now, is this statement correct? If the preservation of offspring is the strongest instinct that governs the honey bee, then why does she remove unsealed larvæ from the cells, to make room for a rich honey harvest? Mr. Otis, of Wisconsin, claims that the strongest instinct of the working bee is the love of storing honey. So it seems the position assumed by Mr. Seay, is at variance with that of Mr. Otis, and one or the other must of necessity be wrong.”

As to being at variance with some eminent *bee*ologist, we have not a doubt that it is so, but you know, Mr. Editor, great men will differ. I deny emphatically that the workers will destroy the unsealed larvæ for the purpose of storing honey. I have never seen any evidence of it among my bees, and should be pleased if some correspondent (if he thinks such is the case) would take the affirmative and give the evidence.

To satisfy himself, that the first and highest law of nature in the honey bee is self preservation and the perpetuation of the species, Querist need only have a fair open contest with a hive of bees. Why do they sting? For self-preservation and the defence or preservation of their colony (species). Injure a single bee in the hive, and the whole colony is instantly exasperated. Cause the honey to run out without injury to any of the bees, and the effect is somewhat different. Tear the comb containing sealed brood, and the bees are at once enraged. And for what purpose? For self-preservation as a colony, in caring for the offspring. Why do they gather honey? For self-preservation and perpetuation of the species.

Is there nothing in all this to demonstrate the fact that the first and highest law of nature in the honey bee is self-preservation and the perpetuation of the species?

If this principle did not pervade the universe, everything would be chaos and confusion. It enters into and becomes the fundamental principle upon which the human family, the animal creation, and the vegetable kingdom have their existence. What causes the mother to care for her infant? It can be nothing less than this. If Querist were hemmed in some corner by an assassin who sought to take his life, and he had power to save himself by killing his antagonist, would he not do it? What causes the animal to care for its young, as the cow for her calf, or the sow for her pigs, or the birds for their unfledged young? What causes the bee to sting when the hive is improperly treated, or the smallest pismire to bite when its tenement is disturbed? You may pass from the human family down through the entire animal creation to the smallest animalculæ, and this (as it were) immutable principle pervades the whole series. Every once living thing that has become extinct as a species upon this earth, failed from some unknown cause, to comply with this grand fundamental principle—*self-preservation and perpetuation of species*.

Querist next says—“Again, is it not a fact that the self-preservation of the matured bees, is far stronger than the love of offspring? Witness, for instance, the destruction of drones during a dearth in the honey harvest?” I do not know whether I understand him here. When I say, honey harvest, I mean a time when there is plenty of honey to be found by the bees in flowers, honey dews, &c. Webster’s unabridged gives the meaning of dearth as “scarcity, want, need, famine.” These two terms then stand in direct opposition to each other. A honey dearth within a honey harvest is an utter impossibility. It implies two distinct terms, not both existing at one time, as a man within a man, or a horse within a horse. Language seems here to have betrayed Querist over to my side of the argument. It is true that the workers do destroy the unhatched drone brood in time of dearth. But why do they do it? It is in strict obedience and conformity to this alleged first law of nature.

Does Querist not know why his bees are so slow about entering their honey boxes, for the purpose of building combs? It is simply this grand

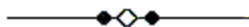
fundamental principle that prompts. It is only because there are supernumerary bees in the hive that a portion of the workers leave the brood and enter the out of-the-way receptacle. The temperature required to produce brood is 70° to 80° Fahrenheit; and the amount of brood produced is governed by the number of mature bees in the hive. If the greatest instinct in workers be to gather honey, why do they not abandon the brood *en masse*, go into the honey boxes, and begin comb-breeding, when the grand flow of honey is to be found in the flowers? Because they would thereby doom the colony to inevitable destruction. Why do not bees enter honey boxes of their own accord, without waiting to be coaxed (as is generally the case) by placing therein small pieces of empty comb? Because their numbers will not permit them to leave the brood. And the same law of instinct, steps in and tells them that the brooding department must be run, whether combs are built and honey collected, or not. Why do they not build combs as readily in honey boxes above the combs containing brood, as they will in an open space below? Because they can thus produce the required temperature of 70° to 80°, and the heat generated below will ascend through the brood combs and bring about the same temperature above also (among the brood), thus accomplishing a double purpose, by virtue of the natural tendency of heat to ascend.

Querist says—"Mr. Seay has much to say about brood chilling." This is true, and I have still more to say about it. It is this—it is brood just hatched, or not more than four days old, that is so easily chilled. This brood is very hard to see in the cells, and bee-keepers are not looking for it to be chilled; but when it becomes so and is lost, without having been seen in that state by the inattentive observer, its destruction is not the less attributable to that cause. Querist says where he lives, "sealed brood is not very likely to become chilled during June and July—the swarming months, and but few bees are necessary to keep it at the proper temperature to mature." We do not know where Querist lives, but we do know that in Iowa in the months of July and August, on replacing our frames after handling them for some time, when the temperature was rather low for those months, we have frequently designated the place in the combs where young brood existed, by piercing the combs in a circle around it, with short stems of timothy grass, and left them there for a day or two that I might be sure to find the exact place and cells again; and, in many cases, on re-examination, I found no

brood in those cells. I have repeatedly made swarms in the Langstroth hive, and afterwards found that the brood, in what I call the first stage, was gone.

J. W. SEAY.

Monroe, Iowa.



[For the American Bee Journal.]

Report of Apiary in 1870.

THE FIELD. The farmers cultivate their fields for produce for the city. They are so frequently broken up that white clover has a poor opportunity for an abundant crop. But little buckwheat is sown. This season none of any consequence within three miles. Fruit blossoms in the spring were unusually abundant.

THE SEASON. The early part of the season was favorable for gathering honey. The breeding apartment of the hives was well stored with brood and honey at the commencement of the white clover harvest. This harvest was, however, shortened by the drouth, and no honey was stored in boxes after the middle of July; and in some cases honey was removed from boxes partly filled.

NUMBER OF COLONIES. I set upon the stand in the spring twenty-three colonies. Of these, three were in old box hives which were broken up when they cast the first swarm, and the hives converted to kindling wood. One of the remaining twenty, from loss of queen or other cause, failed entirely; and a new swarm was introduced to occupy its place. This left nineteen of the old colonies, for giving swarms and surplus honey.

SURPLUS HONEY IN BOXES. I find on adding up the product from my hives, they have given me one thousand and eighty (1,080) pounds of surplus. Perhaps in an ordinary field and poor season I should be content with this; but I think, with the experience of this season and some improvements in my hives, I could do better tried over again.

Of this 1,080 (or to be exact, $1,080\frac{1}{2}$) pounds, five colonies give $625\frac{1}{2}$ pounds, an average of 125 lbs., and $74\frac{3}{4}$ lbs. more than half of the whole surplus. One of the five best gave one hundred and ninety-eight and a half ($198\frac{1}{2}$) pounds.

I attribute this success of my best colonies to the following causes:

1. A full force of workers at the commencement of the season. To secure this, I fed them two or three pounds of syrup, when first placed upon the stand early in March.

2. This gave them from one to three weeks start of the others, in commencing work in the surplus boxes.

3. I think, further, one cause of such force of workers was a most prolific queen. Twelve boxes of six pounds capacity are now almost full of bees, though without honey or comb, except one or two.

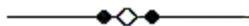
4. But this great number of workers, and early filling the hives with bees, would not have given the surplus had they not been satisfied not to swarm. With the purpose to swarm and preparation for it, they would have given an early swarm, followed by one, two, or three after-swarms probably; and the 198 lbs. of surplus have been placed in other hives in the shape of arrangements and stores for wintering one, two, or three new colonies of bees.

In my experiments with bees, I have generally found a loss of two weeks time in preparation for swarming, in which little or no surplus honey is stored—the great body of the workers clustering out in idleness. Or if boxes were furnished them and filled with bees, I have been disappointed on the swarm leaving the box empty of bees, to find it entirely destitute of honey.

Although my advanced age and infirmities moderate my ambition in the new business of bee-keeping, and so limit my experiments that I have never tried to increase my stock by artificial swarming, I have no doubt but the greatest success in the business can only be secured by the use of non-swarming hives and artificial swarming. Overstocking the honey-field is, in my settled conviction, the great obstacle in the way of satisfactory success. This makes it necessary to have the entire control of the increase of colonies, to limit their number to the capacity of the field. I hope to do better another season, from knowledge gained by the experiments of the past.

JASPER HAZEN.

Albany, N.Y., Aug. 12, 1870.



[For the American Bee Journal.]

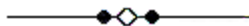
Four-Banded Bees.

Mr. Alley says, in the last number of the Journal, that Mr. Briggs “may bet a high figure that no worker bee in this country ever showed four bands.” I beg respectfully to differ from him, having a queen now in my possession which produces bees that plainly show *four* bands, when filled with honey.

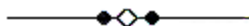
I noticed this before seeing anything about four banded Italians, in any publication. It is true, that the Baroness Von Berlepsch wrote me early in the spring that Dzierzon was selling such queens, but that was the only time that I had heard of them. The queen mentioned above was raised by me last season, and is not purely fertilized, as many of her bees show only *one* band.

DANIEL M. WORTHINGTON.

St. Dennis, Md., Sept. 5, 1870.



A bee-hive is a school of loyalty and filial love.



[For the American Bee Journal.]

Novice.

DEAR BEE JOURNAL:—Just hear the good news,—our bees are again at work! Not, indeed, at the rate of ten or fifteen pounds per day, as in June last; but they are really at work at this date, September 9th.

We had been building some more “air castles,” and had talked of another yield of honey in August and September. After waiting some time, and watching and weighing a hive without any increase, we at last began to perceive a gain in weight, first of half a pound, then a whole one, and yesterday a stock of *Italians* gained two pounds and a half, which was enough to make us toss up our hat and almost embrace the little yellow pets (with judicious gentleness, of course).

A neighbor says the way we follow the bees across fields and through woods, and delve into the subject and remove obstructions, it is no wonder they get honey if it be on the face of the earth—and perhaps that is so.

But, look here, my dear reader, did you understand us to say that our bees were building *combs*? Not at all; “nary” comb will they build, with a few exceptions, and certainly none in those old-fashioned traps called boxes. It is this way. Where there are empty combs right above the brood, they will fill them with honey; as, for instance, in the upper story of the Langstroth hive. But they seldom put any honey in combs very far to one side; and hives that are full, or nearly so, do not increase in weight at all. So you see it all depends on having plenty of *empty* combs. We really think a few more just now would be worth a dollar apiece to us. A little feeding given just right will induce comb building, but we think not so as to pay.

The one stock that we weighed all through the season has now given us three hundred and thirty (330) pounds; and had it not been for replacing their queen, they would have done much better. Their new queen is nearly a black one, and so, also, are her workers; and, by the way, Mr. Editor, here lies a trouble. In slicing the heads off of all our drone brood this summer,

we increased our yield of honey, which was all right. But we increased the yield also of new queens that produce black workers, or at least so nearly black that we have resolved to purchase twenty-five pure queens, to replace all that are not fully up to our ideas. It is true we might raise them, but at the prices at which they are now offered, we begin to think we had rather raise honey, and let some one who has more time or likes the bother better, raise queens. In making new swarms we have no trouble; but in raising surplus queens to replace others, etc., we have not made it go to suit us. We have made some experiments in artificial fertilization this fall, but have not succeeded. Queen nurseries and hatching queens in cages have also been an “unsuccessful bother” to us. We know we are but a poor novice, and should not expect to succeed always, but it does seem as if queens that do not lay, are rather a risky property to meddle with.

But there is one thing we do like, and find it a real pleasure, namely, to keep a *record*. Thus, we found sixty-five stocks too many to remember all about, so we got a blank book with 150 pages (bear in mind it is a good idea to have a few extra pages, even if you are sure you *never will* want to use them). No. 1 hive is on page 1, No. 2 on page 2, and so on to the end of the chapter. Each page tells when the queen of the hive it refers to was hatched, whether pure or not, prolific or not; if weighed, how much honey produced; if queen to be replaced, how and when; and, in short, all about the hive.

Our hives, bees, and combs weigh about thirty pounds each, and before putting them into the house in November, we are going to make every one weigh over fifty pounds, and not more than fifty-five. Some might call twenty five pounds sealed honey (or nearly all sealed) not as well as more; but, as we winter them, we think more would be detrimental, and with us all the rest goes into the melextractor. Were it not for that same melextractor, we fear, or rather *feel sure*, we should not get any surplus honey at all now.

In our last article it read that we had sold all our honey at thirty cents a pound, which was a mistake that crept in somewhere. The honey was sold for thirty cents per pound retail; but the commission, freight, leakage, cost of boxes, labor, etc., made quite a hole in the thirty cents. In regard to saleableness, we have just shipped the last of our three tons, and think that we could sell almost any quantity.

As respects the source of the honey we get now, it is mainly from the same white-flowering plants sent you last fall, which are even thicker here this season than they were then. And, Mr. Editor, we really think that the more bees there are kept, the more honey plants will grow; for every blossom is most surely fertilized, and the result must be more and better seed.

For the first four years that we kept bees, we never found the hives to gain in weight after the first of August; and then we had only from four or five to twenty stocks. Sixty-five colonies is certainly nothing like overstocking, and we have no fear that one hundred would be in any danger if *well taken care of*.

We have found our bees also working so briskly, on what we call fireweed and common golden rod, that we have labelled the honey from

AUTUMN WILD FLOWERS.

It is dark and thick, but has a very pleasant flavor, something like humble-bee honey, as we mentioned last fall, and very different from either clover or basswood honey.

We have had no buckwheat nearer than two and a half miles, and we followed the bees one morning all the way there, as our wild flowers were not then in blossom. We think we can afford, next year, to give farmers within one and a half miles of us, a dollar per acre to raise buckwheat. It is true it might prove a failure, but we are used to failures occasionally.

Many thanks to Mr. Tillinghast, on page 63, and also to yourself, Mr. Editor. When we commenced here with bees, our locality certainly was called poor. Bees had ceased to pay, and were dying out; and had we not been so much discouraged by what bee-keepers told us, we should probably have commenced sooner. One man purchased a hundred stocks, but utterly played out the first year. Black bees are now increasing around us at quite a brisk rate; but that is about all they do.

Mr. Tillinghast says that amount of honey (5,000), in the time, in his locality, "is simply impossible." We think he would have done better to have said, *in his opinion*. We poor mortals very often have a very imperfect

idea of what is possible. After the account was given in our county paper, that our bees were bringing in two hundred pounds of honey per day, and that one stock alone gathered forty-three pounds in three days, it was pronounced utterly impossible; and that if those who told it would consider, they would see that *it could not be!* And we were obliged to invite them publicly to come down and sit by one of our hives all day, weighing it at intervals, if nothing else would convince them, before they were still.

Counting the number of flower heads that a bee visits is a new idea to us; but we cannot think our bees visit more than a dozen certainly. One day in June, when we examined the red clover, we should think a bee would get a fair load from a single blossom; and many of them were working in the red clover at the time. The number stated seems as though the printer had made a mistake with the figures. Nearly ten blossoms in a minute for a whole hour, and not more than a load then! We agree that must be poor pasturage.

Nearly every year since we have kept bees has been called, by more or less unsuccessful ones, the “poorest” season ever known; yet, so far as honey is concerned, all we ask is—more *just like them*.

The only plant we have ever cultivated for bees is the Alsike clover, of which we have about half an acre, sown last spring on the snow, and which has bloomed quite profusely for the last six weeks, but is now nearly gone. We think our bees kept at least one sentinel to the *square foot* of it, to watch for the honey as it collected.

We had a visitor the other day (in fact, we have visitors by the score, and we are ashamed to say, to our sorrow sometimes). Well, this one for a while did not think proper to inform us whether he kept bees on the “brimstone plan” and came to convince us it was the best way, or whether he was the Editor of the BEE JOURNAL himself (of the latter we were very sure, as we think we should know *him* anywhere); but eventually he taught us some things, and we hope he learned some things from us. His visit did not last quite twenty-four hours, but he really made us feel quite lonely, for more than that length of time after he was gone. One simple thing, that Gallup has often said before, but we did not believe it, our visitor convinced us of—namely, that rotten wood is ahead of all tobacco, rags, or anything else, for subduing bees, especially hybrids, who will sometimes “fight till death” when

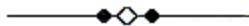
tobacco is used, but would turn around and go down between the frames “without ever a word” under the influence of rotten wood smoke. But don’t do as we did next day after he left us, and drop fire into the saw-dust. We burnt up a heavy two-story Langstroth of Italians before we discovered the muss, and the stream of melted wax and smoking honey that ran out in lava-like channels was a warning to all Novices.

And then we had some robbing at OUR house. We got about half a dozen frames of empty comb hastily put in a new hive, and removed the burnt one, and got the bees to bringing in the honey that had run out (they wouldn’t eat melted wax); but before they had got it all done, there arose an “onpleasantness” as to *ownership* that finally mixed itself into a grand jubilee, in spite of Novice. The burnt hive is patched up, and the combs and bees are back into it, minus their queen, about forty pounds of honey, and ten frames of comb of such evenness and beauty, that some one (who wanted to pick a fuss) said we thought more of them than of our wife and family.

Our visitor aforementioned says he has never written but one article on bees, and we think that so richly deserves a place in the Journal, that we mail it to you.

And now, Mr. Editor, we would say before closing, that in our humble opinion, the results we have achieved this year, are no nearer what *may be* done in scientific bee-culture, than the old brimstone way is to our present method, and humbly beg to be still considered a

NOVICE.



[For the American Bee Journal.]

Bee-Culture in Cities.

MR. EDITOR:—According to promise I will try to answer the queries so often put in the JOURNAL:—“Are bees profitable?” and “Can bees be kept in cities?”

I have kept bees for the last three years on the roof of a two-story house in the city of Cincinnati, having kept bees before, when living on a farm. We did then about as well with them, as our neighbors did who also kept bees; but we were without the aid of the BEE JOURNAL, and kept our bees in common box hives—hence our doings could hardly be called bee-keeping.

Three years ago we took to the city the last hive which the moths had left us, built a platform on the roof of the house, and placed the hive thereon. It threw off a swarm in June following, and gave us some honey. In the fall I introduced an Italian queen in each colony. Two years ago I subscribed for the AMERICAN BEE JOURNAL, and transferred my bees into Langstroth hives. A year ago last spring I entered on the campaign with five colonies of bees—the two Italians in Langstroth hives, and three in Townley hives, having bought the latter. They produced during the season nearly five hundred pounds of honey, all in small frames weighing from one pound to one and a half pounds each; and the fall found me in possession of fifteen strong stands of bees, most of them Italians. On the fourth of June, 1869, I hived two second swarms, clustered together, from two of the Townley hives. After giving them an Italian queen and a full set of empty combs, they produced for me 138 lbs. of honey, the same season.

Last spring I had a first-rate honey slinger made by a brother bee-keeper in this city, and commenced the season with twenty colonies—fourteen of which were Italians or hybrid. As the bees commenced storing honey very early, my expectations were quite flattering, though I did not obtain as much honey as I anticipated. Several mistakes which I happened to make, account for this, in part; but my honey-harvest is respectable still. Here is a statement of it:

384 lbs. of honey in frames.
1,350 " machine strained honey.
23 " beeswax.

As beeswax sells at the same price here that honey does, we may count it with the rest, and thus we have 1,757 lbs. as the product of twenty hives of bees in the city of Cincinnati. This certainly speaks well for our Italian bees, and for bee-keeping in a large city. My black bees have done well, but I think my Italians have given me nearly twice as much honey. Every one of my twenty colonies is now strong.

I was induced last month to make four more swarms, by taking from each hive about two frames with brood, honey, and adhering bees, and giving an Italian queen to each swarm. I have thus twenty-four Italian stands of bees, in a No. 1 condition.

Last year I wintered my bees on their summer stands, by leaving the honey board in its proper place and covering it with about half a dozen coffee bags or pieces of old carpet. I placed a smooth bag next to the board, to cover well the openings. This plan did very well. I did not lose a single colony, and intend to winter them the same way this year. In the earlier part of the winter I lost a great many bees, for the reason that I had neglected to cut winter passages through the combs. This having been done afterward, on the first mild day we had, my bees then got along first-rate. Before this was done, I sometimes found hundreds of bees dead in the cells on the outside of combs which separated them from the cluster—showing clearly the necessity of winter passages. Most of those parts of combs had already a putrid smell, and I thought it best to cut them out.

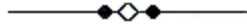
I have seen it stated several times that bees get irritated by tobacco smoke, and are more apt to sting for several days afterwards. This may be true of the black bees. They will bother me sometimes, in spite of my cigar. But I think those assertions are only made by non-smokers. All I want is a cigar, and I will open every one of my hives, take out every frame, and replace it every day for a week successively, without finding my bees any more angry at the end than they were at the beginning.

I learned how to open a hive from Mr. Gallup, through one of the numbers of our BEE JOURNAL. I hardly blow any smoke at the bees, but over them; and I keep my cigar in the mouth, while Mr. Gallup keeps his pan with sawdust by his side, until the proper time arrives for the application of a little smoke. I think there are no more peaceable hives than mine in the country.

Now, Mr. Editor, I do not want to exhaust your patience, and wish you to make use of this, or of such portions only, as you may think proper.

CHARLES F. MUTH.

Cincinnati, Ohio, August 16, 1870.



[For the American Bee Journal.]

The Looking Glass Again.

On page 67 of the last number of the BEE JOURNAL, Ignoramus criticises my article on page 34 in regard to the looking glass, and says the glass has been tried three times this year to his knowledge, and three swarms of bees secured. But he gives us the particulars in only one case, and then guesses at my reply, which is perhaps correct; or the swarm may have had two or more young queens, and a small portion with one queen settled on one tree, while two or more queens with the larger portion of the swarm settled on another. After a few minutes, all these latter queens may have been simultaneously killed, and then the bees went to the other tree and joined the small portion with the one queen. As to the bees coming down to the ground, that is often the case. When a swarm issues, the bees are so full of honey that it is difficult for them to fly, and they often light to rest. I have often had swarms to settle in three or four places, though they had but one queen, remain for ten or fifteen minutes, and then all join the cluster with the queen. Just so with the old woman's bees. They may have just been in the act of going to join the cluster with the queen, when she saw them.

Ignoramus also tells us how to secure swarms with a *knot*. Well, sir, I have never tried the *knot*, but I have tried the *mullein* tops tied in a bunch and attached to a pole, &c., and also a piece of old black comb attached to the under side of an inverted bottom board swung to a pole, with cord and pulley, to raise and lower, as the bees would rise or fall. But after trying both for a whole season, when I had more than a hundred swarms to issue without a bee lighting on either, I gave it up as a failure. I think it likely his knot theory will answer very well in a prairie country, or any place where there is nothing for the bees to light on. But where they are surrounded with as many shady fruit trees as mine are, they will mostly select a leafy branch to settle on. When I allowed my bees to swarm naturally, I had two-thirds of the swarms, or more, to settle on the under side of my grape arbor; which proves that they prefer a cool shady place to a bare pole with a knot on it.

Ignoramus says I remind him of an old Dutch lady, &c. Well, sir, I am like the Dutch in *one* respect; that is, I am in favor of progress; but I am not like the old Dutch lady you refer to, for I was persuaded by your suggestion to look again into the glass and well. Yesterday was a clear, bright sunshiny day. I took a glass some fifteen inches square, and just as Ignoramus said, I saw different from what I did on the other occasion. I saw the water in the well and my own *pretty* face in the glass—nothing more. I am now ready to try any other experiment that Ignoramus may suggest; but my opinion is, the better plan will be to throw aside the glass and make artificial swarms. Then there is no danger of any going off, besides being the fastest way of increasing bees, when the operator understands the principle well. But had I been wholly like the Dutch lady, I should never have succeeded in making artificial swarms. In my first efforts, I ruined dozens of swarms before I succeeded.

I am aware there is much yet to learn about bees, and my motto is to try and try again. So come along, Mr. Ignoramus, with your suggestions. If you do not teach me anything, you perhaps instruct somebody else, as there are many new beginners that read the Journal; and the Journal is the place to receive and impart bee knowledge.

H. NESBIT.

Cynthiana, Ky., Sept. 6, 1870.



[For the American Bee Journal.]

Great Number of Queen Cells and Queens Secured from One Hive.

MR. EDITOR:—In volume 2, number 9, of the American Bee Journal, Mr. A. Grimm gives a case, under the above caption, of forty-three queen cells on one frame of comb. I have had two similar cases this season. The first one had twenty-eight cells on one frame; the other had forty-seven cells on one, and five on an adjoining frame—making fifty-two cells at one time, in one hive.

Early in the spring I experienced the greatest difficulty in getting my bees to start queen cells in full stocks. Having an extra choice queen, which I intended to raise from exclusively for the present; and not being willing to risk the loss of her in moving her from one stock to another, I adopted a different course. (By the way, I always start queen cells in full stocks—never in small nuclei.) I removed the hybrid queens from three strong stocks in succession, and in five days after their removal, I cut all the cells then started, and gave each stand a frame of brood and eggs from the choice stock. On opening those stands a few days after, to see what number of queen cells they had started, I was doomed to disappointment. The first one had only three cells, and two of these were built too close together to be separated. The other two stands did very little better. Getting tired of this slow process, I removed the queen from another strong hybrid stock; then exchanged the whole of the brood combs with the choice stock, brushing off the bees into their own hive. In this way I got some sixteen cells.

On the 6th of June two very large swarms got together. I divided and equalised them, and thinking each had a queen, I left them and went to other work. One of the queen's wings being cropped, I had put her on the cluster before the other swarm issued—the two stands sat about a rod apart. About an hour after this one of the stands became restless, the bees flying out and in, but neither going back to the old stand, nor to the one I had just separated them from; nor settling, either, except on the tops of the weeds and grass, two rods below the two stands, and under the limb they had

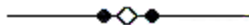
swarmed on. It then occurred to me that the cropped queen might have dropped in the grass, and I started to look for her. But what a sight presented itself to my eyes—a great, big, long snake! No, not a snake, but a bee procession, a rod long and from three to five inches wide, travelling on foot, through the grass and weeds, to the nearest stand, headed by her majesty—who just entered the hive before I could seize and secure her. This was the stand from which I had just separated them an hour before. I then had my work to do over again, which I did in a few minutes, but got both queens in one hive, though I did not then know it. I had watched closely, and saw only one queen enter. By this time other swarms claimed my attention, so that I hastily took a frame of brood from another stand, and gave it to the one I was not certain had a queen—intending to give them one as soon as I ascertained it needed one. They went to work, as though all was right; and I paid no more attention to them till the second day after, when I opened the hive to examine. I found they were building straight and nice worker comb. I did not then raise the frame of brood, as the nice worker comb satisfied me that they had a queen; that is, according to the authority of book authors and others, that bees will never build worker comb without the presence of a queen. But here is an exception; and I have in my practice come across many exceptions to general rules, where bees are concerned. On the 19th this stand swarmed, and taking advantage of my dislike to work on Sundays, went to parts unknown, though I saw them go. I was then engaged in hiving four others, and they refused to await their turn to be waited on. Next morning early, I raised the brood comb already mentioned, and secured seventeen fine queens, counting twenty-eight perfect cells in all! The hive was about filled with comb, but only about one-third was drone comb—the rest being worker comb. Nothing ever puzzled me more than this case. I cannot account for it without going counter to the established rules, that bees without a queen will build drone comb exclusively. But, as I said above, this swarm was extra large, and having a frame of brood given them at the start, may have taken a notion to divide again, and so built worker comb while raising the queen cells. Or, will some one say the old queen was present. Well, if she was, why did the bees build about one-third drone comb? Will some one give us a similar case—such as a newly hived large swarm starting queen cells at once, while they have a queen. I am almost positively certain that they had no queen; yet there is much about the case that bothers or puzzles me. *A good job for Gallup!*

On the 27th of July, I removed a queen from a strong nucleus, to send her off. The nucleus hive was $12 \times 12 \times 18$ inches, with three frames and partition board. It had been started with two frames, but an empty frame was afterward inserted in the middle, to give the bees more room to work. This frame they had filled out to within two inches of the bottom. I had disturbed the nucleus a few days before, to stimulate the queen to lay before removing her. In six days after her removal, on opening the nucleus, I found and counted forty-seven perfect cells, but saw none on either of the other frames; yet, while removing the cells on the 10th day, I found five more on one of the adjoining frames—making fifty-two (52) in all!

In conclusion, let me add that this has been a poor season here. I will get only about 500 pounds of honey, to Novice's 5,000. Hope he has filled his cistern by this time. But here I must close, as I have already wearied the patience of your readers.

R. M. ARGO.

Lowell, Ky., Aug. 12, 1870.



[For the American Bee Journal.]

Bees in Iowa.

When the spring opened, it found me well prepared with very large colonies; but while they seemed to be doing all they could and working hard all the time, they used up all their stores, and I had to give the larger ones honey in the comb stored last year. Then while the fruit trees bloomed profusely, and when white clover had been in blossom a month, my bees had not capped—even in the largest colonies—a pound of honey, much less built any comb. Otherwise they did well.

In the winter I had thirty-five stocks. In January I smothered one, and in April three proved queenless, and two others were robbed; thus leaving me with twenty-nine. Since then I killed a drone layer, and in another hive the queen died and the bees had mostly *gone up* before I discovered their loss. I gave them queen cells, and as they hatched out a week ago, tomorrow I shall examine all my new swarms and see if any failed to secure a fertile queen or lost theirs. Thus you see I was reduced virtually to only twenty-seven stocks. Now, I have thirty-eight, and, with the exception of one, all are very populous.

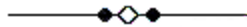
As we have not had any rain here this spring, except one or two slight sprinklings, we are now threatened with drouth. Heavy dews and a clouded sky have saved us so far, but have kept the bees from flying a great deal. I shall not increase my stock any more till it rains, or honey becomes plenty again. From the hive that I have raising queen cells, I secured fifty in three weeks.

On the 11th of this month (June) I received an Italian queen from Mr. Charles Dadant. I was disappointed when I first saw her, as I had formed the opinion that the Italians were a larger bee than the blacks; yet there is not a worker in my hives that is not larger than those that came with the queen, and I am positive that I have black queens that are almost three times as heavy or large as the Italian queen I received. But the Italian is quicker than lightning and the workers are on guard the first in the morning and the

last at night. I introduced her to the colony raising queen cells last Monday morning, giving the black queen to a queenless colony. I examined the hive containing the Italian this morning, and find that the swarming impulse is still on them, though the introduced queen is of this year's raising, as Mr. Dadant says, "she was born this year, 1870." On examination, I found twenty-five queen cells in the hive, ready for the egg, if the eggs are not already in them. It was too early and still too dark, being "before sun rise," for me to make out if any eggs were laid in the cells. When I removed the black queen, I destroyed even the old queen cell foundations, so you see my mode is not theory but fact. As fast as the queen cells are capped, I shall remove a black queen from a colony and give it two queen cells, to make sure of one, till all have been changed to Italians. Next year, when I shall have none but Italian drones, I will easily secure pure Italian stock.

J. M. PRICE.

Buffalo Grove, Iowa, June 20.



[For the American Bee Journal.]

The Honey Season in Jasper County, Iowa.

MR. EDITOR:—This has been a somewhat poor honey season in this locality, owing to the dry weather. The month of March was pleasant and warm for the season. At the close of the month there was brood in the combs in most colonies. April was less favorable. The month was cold, and at its close there was less brood in many colonies, than there was at its commencement. May was warmer again, and the bees commenced gathering pollen early in the month. Breeding was extensively resumed, and towards the last of the month, the bees stored some honey. Most of the hives were strong and apparently in good condition to be divided; yet a division at this time, or in fact at any time during the season, would have proved injurious to many, if not entirely ruinous to some of the divided colonies. Honey gathering ceased with the failure of the fruit blossoms. No more honey was gathered until the last of June. Through the middle of that month most stocks were nearly destitute of honey, and the drones in most colonies were killed off. The slaughter was pretty general. About the last of June the bees commenced gathering honey again, and for nearly three weeks it was stored quite freely. Towards the end of July the honey harvest ceased, and from that time till within the last few days bees gathered no honey.

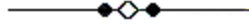
As a whole, the season has been a poor one. Very few stocks swarmed—especially of natives. The Italians have done better, those at least that were rightly managed. In the spring I placed twenty-eight (28) colonies on their stands, all of which had been wintered in a dark cellar. These I have doubled by artificial swarming, except three natural ones.

I drew and started up twenty-five (25) nuclei, for queen raising purposes, and *kept them up*. This I have done, while my neighbors did not get either swarms or honey; yet I do not think I have any colonies but what will be in good condition for wintering, at the close of the season.

Enclosed please find four dollars, for which send two copies of your valuable Journal, addressed as below. Success to the Journal.

J. W. SEAY.

Monroe, Iowa.



[For the American Bee Journal.]

Introduction of Unimpregnated Queens.

That the introduction of unfecundated queens should be so often spoken of, and that too by some of our experienced bee-keepers, as a matter of much difficulty, is a question to me almost incomprehensible. In the hands of the inexperienced, or of those ignorant of the first principles of success, a few failures ought not to be wondered at. But for those having a knowledge of the prerequisites for the acceptance of a stranger queen by a colony of bees, to talk of the safe introduction of unimpregnated queens, as an act of uncertainty, induces me to believe that they have either not experimented at all on this part of practical bee-culture, or else did so to little profit.

If it be true, as has been asserted time and again in the BEE JOURNAL, that the only means the bees have of recognizing strangers, is by the sense of smell, it stands to reason that, if a stranger queen be confined in a hive long enough to acquire the scent of the hive, the bees will immediately accept her as their own, especially if they have no young queens in process of rearing.

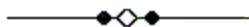
Acting upon this principle the past summer, I confined my young queens in small wire cages, and inserted them as near as I could in the centre of the hive; at the same time taking the precaution to provide them with food during their confinement. The result was that out of a goodly number of unimpregnated queens, introduced in swarming time, not one was lost. We have also succeeded admirably in introducing them, by scenting both queen and bees with some liquid having a peculiar scent. By either method, we regard the safe introduction of a queen bee, whether fertile or not, as a matter of certainty: where the queens themselves are kept from starving by proper feeding.

We permitted natural swarming to some extent this summer, in order to get hardy and prolific queens. As we will break up a number of after-swarms this fall, which were unfortunate in coming late, we shall be able to furnish

some who prefer tested queens to all others, with a number of finely colored queens raised in natural swarms, cheap for cash.

J. L. McLEAN.

Richmond, Jeff. Co., Ohio.



[For the American Bee Journal.]

Introducing Queens.

As an introducer of queens I have not been always successful. In several cases, after two or three days caging, the queen has been accepted all right, and within twenty-four hours rejected. I watched one of these cases, in which the queen, when liberated from the cage, was caressed by the bees, until by and by one of a different mind (and of a different body, too; for I have noticed the first to attack a queen are the small-bodied fellows) assailed her, and very shortly was joined by others, until a mass imprisoned her.

With Mrs. Tupper's favorite method I have sometimes succeeded, and sometimes failed; but then the fault may have been all my own. I have half drowned bees, queen and all, with diluted honey strongly scented with peppermint, and had the pleasure of seeing the drunken fools fondle her as if they had always known her; and then some one of the number, not fully saturated, would attack her.

Latterly, I have taken a different plan, and one which, according to all the authorities ought uniformly to fail; but which, so far, has uniformly succeeded here. It is simply this:

Wait until the bees have started queen cells. Then, without any preparation whatever, put any queen, fertile or unfertile, directly on the comb, among the bees. That is all.

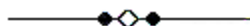
It may be that I shall fail the very next time; but, until I do fail, I shall continue to practice this plan. I give it to the Journal, in hopes that some one else, having a queen or queens of no value, will give it a trial. I have not tried it long enough to consider it a settled thing; but shall report to the Journal the first case of failure. Let me relate a case of success:

August 1st, I put into an empty hive, No. 15, one frame containing some honey and a very few cells of *sealed* brood. I put into this hive a young queen that had just commenced laying, and set the hive in place of one

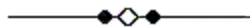
containing a strong colony. Of course the empty hive received all the flying force of the strong colony. On the next day they had destroyed the queen. I then took a queen two or three years old, covered her with honey completely, and dropped her on the frames. She was received all right. Next day, August 3d, I killed this queen and introduced a *young* one in exactly the same manner. She was promptly imprisoned, and I released and caged her. August 5th, this queen having been caged two days, is still refused. August 6th, she is caressed by some of the bees, but others imprison her. I then gave her to a full colony, No. 1, which was queenless and had queen cells started, some of which were sealed. Placing her directly on the comb, without caging, she was kindly received and soon commenced laying. I then took from No. 1, the frame with queen cells, and gave it to No. 13. Three days later, August 9th, I gave to No. 15, an unfertile queen three days old, placing her directly on the comb. On the same day I gave another full colony, having queen cells only a day or two old, an unfertile queen three days old. Being out of the State I did not see them again till August 22d, when I found both queens laying.

C. C. MILLER.

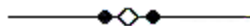
Marengo, Ill., Aug. 30, 1870.



The smell of their own poison produces a very irritating effect upon bees. A small portion offered to them on a stick, will excite their anger.



After a swarm of bees is once lodged in their new hive, they ought by all means be allowed to carry on their operations, for some time, without interruption.



[For the American Bee Journal.]

Bee-culture, Honey Products, Honey Markets, &c.

MR. EDITOR:—I herewith send you two dollars as a further fee of incorporation in the bee family. I have profited well by it this year. I was absent on a tour in Europe last spring. On my return I found my bees in poor condition. Two colonies had died from dysentery or the warmth of the bee cellar; and of the remaining sixteen stocks, two were very weak, with some others in prime order. I had but two Italian stocks left. As far as my experience goes, I must give three cheers for the Italians. The earliest natural swarm I got here from blacks was on the 17th of June. This year my first Italian swarm came off on the 13th of May. The parent stock was a good one, though I cannot set it down as my best in number of bees. I had black colonies that were more populous. As for this Italian, it yielded me fourteen natural swarms, four of which left for the woods and the remaining ten are in extra condition for wintering. The parent hive and the first swarm are the heaviest stocks in my apiary. I shall Italianize all my colonies this fall. No man will ever persuade me that black bees are as good. I shall always consider such men as jealous or prejudiced. The advantages derived from Italian bees are well worth paying for—their early swarming and their rapid breeding are sufficient compensation. The color of the queen, too, is a great advantage when looking for her in the crowd on the comb, and her superior fertility is an unquestionable fact. The fourth swarm came off in May. It was small; but as it had a beautiful Italian queen, I put it in a box hive, and today it has nearly filled a twenty pound box. The season from the beginning of May to the middle of July was very good. My hives were so full of honey that no empty cells were to be seen. I have brought up the number of my colonies to forty-five, and four swarms left for the woods; and thus far I have sold seven hundred (700) pounds of honey.

According to the Report of the Commissioner of Agriculture, there are between 70,000 and 100,000 bee-keepers in this country. If so, the number who subscribe for the BEE JOURNAL is comparatively small. Why is this so? According to my observation and experience there are two reasons. First, because the population of this republic is largely composed of emigrants

from all nations, and although they and their immediate descendants may speak and understand English, yet they are not able to read or write it readily. Every one sticks more or less to his native language, and prefers reading newspapers printed in that language, because he understands it best. The second reason or cause is jealousy. It is a fact well known to every bee-keeper away from large cities, that the sale of honey is very slow in small cities and towns; and it is often impossible to sell at a remunerating price. Thus, for instance, Green Bay is a city of 8,000 inhabitants; yet one bee-keeper with 100 hives can fully supply the annual market of that city in a good year. It is of vastly more importance to write on this subject and induce an extension of the market demand for honey, than to teach fertilization by one or more drones. Bee keeping is now very profitable—more so than is acknowledged in print; but men have a disposition to keep the thing to themselves. It is very often the case that a bee-keeper instructs his neighbors in the art of managing bees successfully and profitably, and as soon as these are well posted in the business, they become a source of annoyance, contempt, and jealousy to their instructors. This makes it the more necessary to make more extensively known the best honey markets that are now to be found, and any additional outlets and uses for honey that may be opened or devised. In France enormous quantities of honey are used in the fabrication of honey bread, called *pain d'épice*. I wish our friend C. Dadant would give us a receipt how to make the best kind. This might become an American institution as well as a French one. The reputation of this delicacy is world-wide, as well as that of the French wines so much liked here. Vinegar also is said to be of superior quality, when made in a perfect way from honey. I should be glad to obtain some reliable information as to the *best* kind of it. Much honey is spoiled, as many other things are also, by using it when not properly prepared. Let us have the true results of experience. Another matter, not less important, is the preparation of good mead. A bottle of good mead is equal to the best wine; women in confinement use it in preference to wine, and with far more benefit. I think mead can be made as cheap as, or cheaper than whiskey. Good fermented mead ought to be sold in all wine stores for medicinal purposes and other uses. It is used in Belgium extensively as a summer drink.

BEE HOUSE.

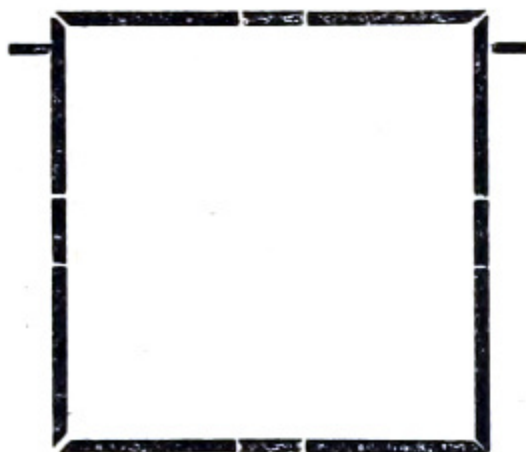
I am going to build me a bee house of cedar logs, twenty feet by sixteen inside, stuffed with one foot of saw-dust; and I wish to know how I can give the greatest amount of ventilation in winter, without light. I want the largest amount of ventilation, combined with the largest amount of darkness; and desire to know where and how to place the ventilators, and of what material these should be made—whether of wood, iron, or lead? If possible, let us have a sketch or side view. Did I not fear that NOVICE was drowned in honey, I would ask him to have the kindness to furnish the information according to his experience. Perhaps we should send in contributions to the editor to offer a premium for a design for the best bee-wintering house, to contain a hundred hives as described above. Bee-wintering is one of the most important points in bee-culture now, and bee-keepers could well afford to contribute towards procuring the best plan of a house.

Now, dear editor, although a passenger in the sleeping car, I am for progress. Thirteen swarms from one—say one brought up to fourteen, is a true American fact. If I had set the fourteen in four hives, with ample space for boxes, it would have been a pity for my blacks to compare results. I drummed out my old hive and first swarm, and cut three pails of honey out of them. Then I returned the bees, and the gaps are again nearly closed. I wish now to say

SOMETHING ON HIVES.

Last year I made me three Price hives according to Vol. IV., page 87. On inspecting my hives, after the bees had been put in, I found in the first one all its frames lodged on one side. To obviate this, I drove small tack-nails on top sideways, to hold the frames at proper distance apart; but this does not do. In lifting out the frames I slightly damaged brood and honey. The second hive was in order, but the combs very uneven. The third had its combs straight every time, impossible to be otherwise down to the middle; but from the middle corners down to the lower corner they were fastened together and all gone astray. Further, the crushing of bees by the honey-board annoyed me much. They are so very heavy and troublesome to handle, that I have broken up the whole concern.

Now, I have constructed a hive on the Gallup pattern, say one foot square, and use twelve frames in it. This is what I like. My combs are as straight as a piece of board, and very easy to handle. I shall stick to it. But, dear editor, I fear I have infringed on some one's patent, and I do not like others to do the thinking, and myself to reap the harvest—which is about as criminal as stealing another man's brains. The question is: whom have I to pay? My frames are made thus:



They hang on a rabbet, suspended by half an inch of iron wire, the thickness of an ordinary lead pencil. They are very easy to take out, and are never gummed fast. Now, do you not think I have infringed the Langstroth principle? If so, please inform me. My frames are three-quarters of an inch thick, and are very strong. I have had much trouble with frames as commonly made, when filled with honey. They are then too weak.

Finally, I have constructed

A HONEY MACHINE

according to Mr. Hubbard's description. I had not the slightest trouble in making it. My can of zinc, eighteen inches in diameter and twenty inches high; cost three dollars. The iron wire cost one dollar, but I had more than enough. The whole cost was less than five dollars. I used the crank of a fanning-mill, to see what effect it would have, but found it too long. I was compelled to turn it with a peg half way down, which is just the thing. I can

turn it as rapidly as wanted—so rapid, indeed, that the larvæ would be thrown out. I shall use no gearing. I found the machine all that could be desired, and only regret that I had it not in June. The queens might have produced some thousands of pets more, if empty cells had been provided for them. Now, something about>

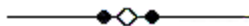
STRONG STOCKS.

NOVICE says if we are well-rooted anywhere it is in *strong stocks*. This, I find, is a very indefinite saying. I wish some one would give me a clear idea of what is meant by the expression *strong stocks*. Is it a large, prime swarm, or a first and a second swarm united, or any swarm well wintered and built up by spring feeding on Gallup's system?

Ah, indeed, N. Woodworth, of Rochester, Wisconsin, on page 47, Vol. VI., has thrown a skunk in the face of the bee family. A skunk cannot stink more than that statement. Surely, he designs to see what effect it will have. Well, the best way is to let the skunk alone. The meanest bee-gum bee-keeper who manages to winter his bees so that they do not all die, has to acknowledge that bee-keeping pays; how much more can one accomplish who knows how to employ skilfully scientific means and methods?

JOSEPH DUFFELER.

Rousseau, Wis., August 26.



[For the American Bee Journal.]

Queen-Breeding for Improvement of Race.

MR. EDITOR:—In the September number of your excellent Journal, page 58, Mr. Alley accuses the writer of “pitching into him.” But I find he can still hold up his head and “pitch” back, as well as raise cheap queens; so he is not badly wounded. But, to be serious, I most sincerely regret that any sentence in my article, in the August number, was so worded that it was thought to be personal. It has been a favorite project with me to see the honey bee improved to its highest possible extent. And even Mr. Alley concedes the principle for which I contend. For, says he, “*I pay the highest prices for my breeding queens, and now have queens of my own raising that I would not sell for fifty dollars.*” This is a higher price than I proposed for such queens, five or six times over. He says he will take my whole lot at my figures, if I have such queens as I describe. I would not like to spare them, Mr. Alley, for I value them as highly as you do *your* best queens!

I do not doubt that every man who gets a queen from Mr. Alley, or from any other man who sends the genuine breed, gets the worth of his money; but what I did mean to say, was, that if a man wishes to get the highest grade of Italians, let him get one that has been raised from the best selected stock, under the eye of an experienced apiarian, and thoroughly tested before she is used as a breeder. Then the buyer will know what he is getting, and would find his purchase cheap at twenty dollars—rather than one that was untested and raised at haphazard, at two dollars and a half.

I repeat—Let the Queen-Raising Brotherhood unite to state these facts fairly and squarely before the world; and let men who believe in sharp practice keep such things out of sight.

I, too, if ever I go into the business again, will sell queens at \$2.50, sending them out as soon as they begin to lay eggs, to any number ordered, guaranteeing that all the workers shall show three yellow bands, when filled with honey. But, if tested and guaranteed as breeders, I would ask ten dollars each. If I was going to commence Italianizing an apiary, I would

send to some responsible man, such as Langstroth, Colvin, Quinby, Gallup, Mrs. Tupper, or Mr. Alley; and in the room of sending \$2.50, I would say, "fix your own price, but send me the best queen you can select!" for I would rather have such a one than four of average untested queens. And putting the seller upon his honor, I think I should get the *best*, where all were good.

Others may differ from me in opinion, yet I have given the public my views honestly.

Mr. George C. Silsby has my thanks for his courteous criticism of my article. Mr. J. E. Pond likewise, though he misapprehends my intention to attack any one but sharpers, who sell for pure Italians what no one, qualified to judge, would call even a good hybrid. I know nothing of Mr. Alley only through his advertisement, and of course knew nothing of the quality of his bees. But while I know nothing of him, I do know men who sent to where it was most convenient and cheapest, and straightway they became queen-breeders, and supplied the country round, in turn, with *genuine queens*. It would take an expert often, to detect a particle of Italian breed in many such colonies that I know of.

In such cases, often, the queen-breeder himself did not know that he was selling a spurious article. I may have been foolish, but I did send to Italy for stock that cost me twenty dollars each, when I could have procured stock from Mr. Langstroth for five dollars each. The same year I procured a queen from Mr. Colvin for fifteen dollars, tested, in preference; and the very next year I sent fifteen dollars to Mr. Langstroth, for a tested and superior queen, when he would have sold me an untested one for half the money. I think still that the money was well invested.

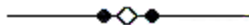
Two years ago I left the personal supervision of queen raising, and a gentleman by the name of J. L. Strong is now conducting the same apiary, at Mount Pleasant, Iowa. He has not been able to supply all his orders this season. My articles were dated from that place; but my residence is at Ottumwa, Iowa, where I am trying to fill the place of pastor of one of the Methodist Episcopal Churches of that city. I have raised just *four* queens this season, one of which was a hybrid. These I have used in making new swarms. I have five colonies here, which still interest me greatly, although

there are not many dollars and cents, as income, in the enterprise, and I take all the profits in honey for my table. So you see I am not a very formidable rival in the trade.

But, in common with the brotherhood, "*bee on the brain*," is a chronic complaint with me, and I never shall recover from it; and every man who talks *bees*, or writes *bees*, or *raises queen bees for \$2.50*, or any other price, has traits that make me regard him as a *brother*. And if I write an occasional article, don't think I am "pitching into" some one, or writing to "show off." Then, further, if you find my articles only half as interesting to you, as yours are to me, I shall be content. In the meantime let us raise no false expectations; but so write that we can put in the hands of the cottager, occupying a few square roods, the means of keeping, in an intelligent manner, from twenty to one hundred colonies that shall bring him as much profit as the owner of a farm reaps from his broad acres.

E. L. BRIGGS.

Ottumwa, Iowa.



[For the American Bee Journal.]

The Economic Hive, and Gallup's.

MR. EDITOR (and some one says that means everybody):—As I receive many letters asking what I think of the Economic Hive, mentioned and described in several numbers of the last volume of the Journal, suppose you allow me to answer them at once through the pages of the Journal. It will save me much trouble, and obviate the necessity of replying to the same questions asked over and over again, by different inquirers. Another matter I would like to speak about. I receive a great many inquiries somewhat like this—“Mr. Gallup, I am a new subscriber to the American Bee Journal.” &c., &c., and asking me for information about such and such articles, or what does such or such a writer mean, &c. Now, gentlemen, I am perfectly willing to answer your questions, but it appears to me that your very best plan would be to send the money to the publisher, and get the back numbers of the Journal. You would certainly get the worth of your money; and then you can understand what the writers mean, better than I can tell you in one short letter.

Well, here I am off the track, as sure as fate. To return; in the first place, the Economic Hive and the hive I use, are (with slight variation) substantially the same. Both can be used in the same manner, in every respect. I have used them with from ten to fifteen frames, but for general use, twelve are sufficient. All it needs is to make the hive wider or narrower, to accommodate more or less frames. In using my hive two story, I make the second story the same depth as the first. My frames hang on small three-cornered cleats instead of on rabbetings; and to make any hive into a second story box, draw the small finishing nails out of the cleats and nail them on again, low enough down to allow one-fourth of an inch space between the upper frames and the lower, without the honey-board. Now, all that is necessary to convert this into two hives, is to move those cleats back to their former places again. In placing this top box on and lowering the cleats, it leaves an inch and a quarter space between the top of the lower frames and the honey-board. Now drive four finishing nails into the sides of the hive, inside, leaving the heads project one-fourth of an inch above the

frames. Then fit in an inch board and let it rest on those projecting nails. This will fill up so much of the vacant space under the honey-board.—In putting on the third story, I make my boxes so as to fit inside the hive, *on the frames*, and do not use the honey-board between the boxes and hive in any case. This third story is only used with very strong stocks.

Once more, I will say that this hive suits me, and can be used for every purpose, in forming nuclei. You can raise four queens in it, as Mr. Truesdell says, and by inserting three division boards you can make it into four small hives. The entrance on the four sides of the hive are all in the bottom board. It can be accommodated to any size of swarm, simply by using the division boards, or not, as the case requires. In short, read what Mr. Truesdell says about the hive, and also what I have previously said about it; and then read what I say in the “Annals of Bee-culture for 1870” (when it comes out) about the best method of having honey stored in combs for market—decidedly the best, in my opinion; better than any glass boxes I ever saw. In such a hive you have one adapted either to a poor honey district, or to a good one. It will accommodate the largest, as well as the smallest swarm you ever saw. It is cheap and simple. Understand, I am not cracking up this hive to make money out of it, for it is not patented, and I have no time to make any to sell.

E. GALLUP.

Orchard, Iowa.



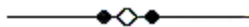
[For the American Bee Journal.]

The Gallup Hive.

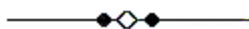
I wonder sometimes how many bee-keepers have tried the Gallup Hive, there being so many other hives that are so highly recommended. I have made and used, now for two seasons, more than a dozen of the Gallup form of hive; and thus far I think it is good for all that Gallup claims for it. Simple in its construction, easily and cheaply made, and for one, I cannot conceive how any hive could be better adapted or more convenient to form nuclei with full sized combs, to raise queens, to equalize bees and stores, build up stocks, exchange combs promiscuously from hive to hive, &c., &c. No trouble about the frames hanging true, and I think I can handle a set of frames in the Gallup form of hive in as short a time as I can in the Langstroth standard; and I am using both. If the several parts of the Gallup hive are correctly made and put in place, it is almost air-tight; and yet any amount of air, whether much or little, can be given and regulated, even to the extent of suspending the hive in mid-air, with top and bottom off, if it were necessary. Its surplus honey arrangement can be made to suit location or fancy. I do not suppose that Novice or Grimm, or some others, would do any better by using the Gallup hive; but my circumstances are very different from theirs. And as it is of the utmost importance to me to use only one kind of hive, I intend to use the Gallup form exclusively as soon as I can, without material loss.

HENRY CRIST.

Lake P. O., O., Sept. 7, 1870.



Those that boast most, fail most, for deeds are tongue-tied.



[For the American Bee Journal.]

Palmer Brothers and the Thomas Hive.

It is due to myself and to Palmer Brothers to say that their article, so greatly in favor of my hive, was written without my knowledge and entirely upon their own responsibility.

While I feel grateful to them for their high opinion of my hive, and the impartial manner in which they have spoken of it, I may be allowed to correct two or three items in the description thereof. They have purchased the territory for these hives before the alterations of which I am about to speak were made.

“Advantage 8th” (see BEE JOURNAL, Vol. VI., No. 2, Aug. 1870.) “There is a passage through the bottom board, covered with wire cloth, through which the bees receive air,” &c. After five years’ experience and experimenting with the hive and the best method of ventilating, I now make the bottom board without any hole through it, preferring instead to put a hole through the rear end board of the hive, about one inch from the bottom, and covered with wire cloth. The hole is an inch and a half in diameter, and allows a circulation of air from front to rear. I consider this the best method of ventilating a hive, and in most, if not all cases, quite sufficient, and especially so with an entrance such as I use in my hive, and with which Palmer Brothers were not acquainted for reasons already stated. I will just say the entrance is so constructed, with a double zinc gauge, that it can be enlarged in a moment of time to half an inch deep and the full width of the hive, and contracted in the same time to half an inch square.

“Advantage 16th. The bottom slants to the front.” It may be made inclined or level, as desired by the builder.

“Advantage 18th. One, two, or four boxes may be used.” Six square boxes, suitable for market, may be used.

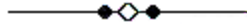
“Disadvantage 3d. The improvements are worse than useless, to one who will not properly use them.” This is true of all frame hives. If a bee-keeper

intends to let his bees die, with no attention on his part, he certainly will save the expense of improvements by setting them in a hollow log.

To those parties who may purchase territory I will send a sample hive, paying all charges to the line. See advertisement, and make an offer.

J. H. THOMAS.

Brooklin, Ontario.



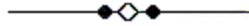
[For the American Bee Journal.]

Bee Cholera.

MR. EDITOR:—I see that many persons have lost their bees by what is called Bee Cholera. I have had some bees die with the same disease. I then took a colony after one half the bees were dead, ventilated the hive well, and carried it into the stove room, and kept it there the space of eight days. It is now a strong colony. I suppose the heat of the room evaporated some of the water in the honey.

B. R. HOPKINS.

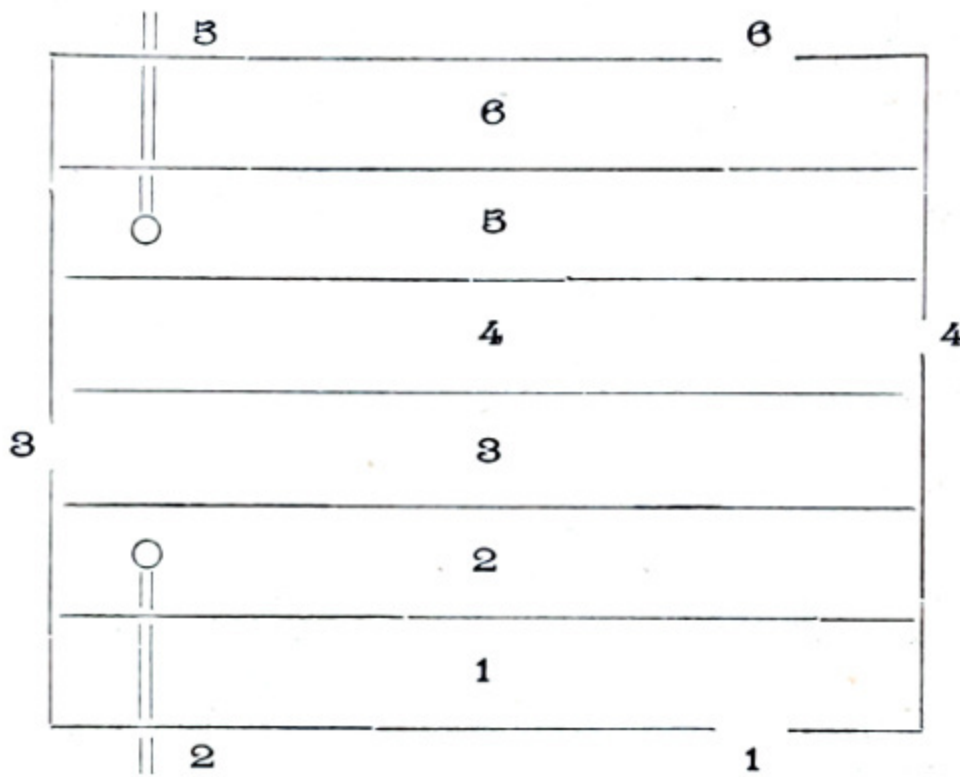
Tyrone, Pa.



[For the American Bee Journal.]

Hive for Nuclei.

The experience of a single season satisfies me well with a hive for nuclei, made by simply taking the ordinary Langstroth hive, separating it into six compartments, and making the entrances face in different directions, in this manner:



Nos. 1 and 6 have the entrances at the back end of the sides, at the upper corner. Nos. 2 and 5 have a hole bored through the bottom, and the bottom board channelled, making the entrances come out underneath the front end of the sides at the *lower* corner. The entrance of No. 3 is in front, at the regular entrance; and No. 4 has an entrance at the back end.

“But will not the queens enter the wrong compartment, on returning from their excursions?” I have raised fifteen or twenty in a hive of this kind, and have never lost any.

Instead of a honey board, a strip of board covers each division separately, so that each nucleus can be examined without disturbing the others.

The ordinary frame is used, and the principal advantage of the hive consists in the mutual warmth gained.

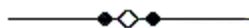
I think it pays to keep reserve queens constantly on hand; and I mean to try whether I cannot winter a few queens in this way.

I have raised some queens by letting the nucleus have brood to start queen cells from; but they have been slow coming to maturity; and after they have laid a few eggs, they are sometimes discarded and a young queen raised from the brood. The trouble seems to be that where queen cells are started by a small cluster of bees, they do not feed the grubs plentifully enough, and when the queen hatches out not a particle of royal jelly is found in the cell. Whereas, when a strong colony raises a queen, the cell will contain a large quantity of jelly after the young queen emerges. To obtain good queens, I take the following plan. I take a frame containing only eggs laid by my best queen, and put it into an empty hive, and set this in the place of a strong colony. Cells will be started and the grubs liberally fed, and as soon as they are sealed over, I cut them out and give them to the nuclei. I then give the hive a laying queen, and two or more frames of sealed brood, according to the time of year, and have a good colony.

I am waiting patiently for NOVICE to invent a machine for making straight worker comb; for as yet I have found no way of securing all worker comb, except to have it built by a weak colony. My bees build some drone comb of very strong, even if their queen is not a month old; and they will build worker comb, *whilst raising queens*, if WEAK ENOUGH.

C. C. MILLER.

Marengo, Ill., Aug. 30, 1870.



[For the American Bee Journal.]

Around among Apiaries.

MR. EDITOR:—As I have been visiting among bee-keeping friends, I will give you a few lines that may interest some of your readers. The season here has been very variable in the yield of honey from the clover blossoms and also from honey dew.

I made a short visit to Hess & Co.'s apiary, some ten miles from Fulton, on the Iowa side of the Mississippi, who have about one hundred and eighty colonies. Their bees did not yield much white clover or basswood honey, but did well on honey dew. The honey from the latter is very dark and sticky, and to most persons is of poor flavor. Their bees did not swarm much this season, though they are surrounded with all the early flowering trees, such as soft maple and hard elm, willow, and all other kinds natural to our soil, alike on the islands, bottoms, and uplands.

I next visited Marvin & Bros., of St. Charles, Ill. Their apiary numbers one hundred and seventy-five to two hundred stocks. Their bees have not done anything to speak of, and from appearance and prospects, they will have to be fed to go through the winter. There was hardly any rain here from the last of March to the last of June. White clover blossomed very little, and Alsike was almost a failure from the drouth. It did not grow tall enough to be cut for seed, where it did come into bloom. But Messrs. Marvin are not discouraged. They think there is a good time coming yet for bees, though it be not this season. They have some of the great Rocky Mountain bee plant growing, but it has not done anything for them since they have had it. It is now in full bloom, yet very seldom a bee lights on it.

I also made a brief call on M. M. Baldrige, the secretary of the great National Bee Hive Company, at St. Charles. His bees will likewise have to be fed, to go safely through the winter, if fall pasturage do not supply sufficient honey for their need. Mr. Baldrige is doing a considerable business in manufacturing honey emptying machines, now that the demand for beehives is over for this year.

I next visited Mr. Thompson, of Geneva. He is young in the bee business, but quite enthusiastic. Although he lost all his bees last winter, he was not discouraged, but tried again this season. Like most new beginners, he increased his stock rather too rapidly, especially in so poor a season as this has proved to be in that section generally. Bees, however, did somewhat better at Geneva than at St. Charles, only two miles away. At Batavia, the same distance below, the bees have done moderately well. Let me remark here that the rains, throughout the West, for the most part went in narrow streaks this season, especially in June, sometimes not over half a mile wide. This accounts for the difference in the condition of colonies in apiaries only a few miles apart.

I called on Mr. Way, at Batavia, and took a look at his bees and honey. He has a good supply of surplus white clover honey on hand, having been fortunate enough to be within the range of one of the seasonable rain streaks. The most of his colonies have honey enough to pass the winter safely, if they should not be able to gather any more. I was told that the good people of Batavia tried to get friend Way's bees expelled from the city limits, as a nuisance, for fear they might possibly sting somebody!

AMONG THE HONEY DEALERS OF CHICAGO.

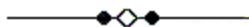
I do not think that the largest honey dealer in Chicago is doing the fair thing by his patrons—that is, if he wishes to do a permanent business and retain his best customers. He would rather buy honey in large boxes and frames, and then cut it into three or four small strips, put it in glass jars, and fill up the jars with inferior strained or Cuba honey. At the same time he discourages the bee-keepers from taking their honey from the combs with the melextractor, for the simple reason, I suppose, that he can make more money by straining the honey himself, as I was told he had a nice steam apparatus for fixing over strained honey.

As to the commission men, there are not many of them to be trusted, as it is seldom that honey is handled with the care it ought to receive; and when it gets to leaking, they sell it for any price they can get, in order to be rid of it.

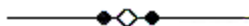
There is a great fault, too, in the manner of shipping it, to have it go through in good shape, as the railroad men do not handle things very carefully. To get the best price from honest dealers, the box honey must be put up in neat, small boxes, weighing not over seven pounds gross; and to get a market established for extracted honey, it should be shipped to some reliable man; and the jars must be labelled with the quality of the honey and the name of the producer. Then the agent can recommend it to his customers, and warrant it pure; and all you have should be shipped exclusively to him. When properly put up, I do not think there is much to be feared from adulteration.

X.

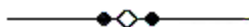
Fulton, Ill., Sept. 5, 1870.



A good swarm of bees, put in a diminutive hive, in a good season, may be compared to a powerful team of horses harnessed to a baby wagon, or a noble fall of water wasted in turning a petty water-wheel.—*Langstroth.*



Narrow minds think nothing right that is above their own capacity.



[For the American Bee Journal.]

Queen Raising.—Experience and Observations.

Too early last spring, I commenced by artificial means to raise queen bees. Using only about a pint of bees, they became chilled during the night, and would cluster in the corner or top of the hive, deserting the larvæ and the unhatched young. This was in March. During the latter part of the month of April, however, I succeeded admirably in hatching them; but two-thirds were lost on their wedding tours.

I had as many as six queen cells which were *to hatch* on a certain day. I was not at home on that day, but returned late in the evening, and on examining No. 1 (a full colony), I found the queen had just emerged, the cap or end of the cell still clinging by a small particle of wax, and the queen on the same frame within a few inches of the cell. No. 2 had also hatched during the day, appearing to be a few hours older. No. 3 was then visited, which was in a nucleus, and I found only two worker bees in the hive,—the queen cell being still perfect. I had the evening before given this nucleus some strained honey, in a bungling manner, and did not contract the entrance of the hive as I should have done, and they were robbed. My wife, early in the morning, noticed unusual activity at this hive. The little family, I suppose, had helped to remove their limited stores to the hives of the robbers, and taken up their abode there, as usually occurs in such cases. But, to return to our queen cell, I removed it carefully and opened the end of it, when, to my surprise, out crawled the queen on my hand. Some honey was given to her, and in a few minutes she was quite lively. She was then introduced to a queenless colony, and was well received; but was lost on going out on the eighth day. No. 4 was not examined until the next day, when a nice Italian queen was moving amongst the workers; with as much dignity as belongs to one not yet having attained her majority. After an interval of about three days, I examined the hive and saw the queen every day until about the eighth, when late in the evening, after sunset, on examination I found she was gone. On closing the hive the bees came running out and showed all the signs of having recently lost their queen, such as are often seen; and kept up that distressing search by crawling over the hive and on the ground in its

immediate vicinity until after dark. The hive was again examined with great scrutiny on the following morning, and she was not there. At eleven o'clock a natural first swarm issued from a hive of native brown bees in the apiary, and after flying around five minutes, clustered on the stem and at the root of a cherry tree. I proceeded to hive them, and when half the swarm had passed into the hive, I saw the black queen march in. Only a few minutes more elapsed before all the bees had gone in, except a little ball or lump the size of a partridge egg near the root of the tree. I stirred them up with a stick, thinking they were not cognizant of the fact that their queen had gone in and the house was prepared and ready for them; but they had no disposition to disengage themselves. Taking the ball of bees in my hand, I examined them and found they were clumped around my lost Italian queen. I dropped them in a pan of water, when every one let go its hold, and the queen was free and apparently unharmed. I returned her from whence she came, and in a few minutes the grieved family were buzzing their joyful wings at her return. In a subsequent examination on that day, she was crushed between two frames. The question arises, how she came to be with this native colony? I have my surmises, but will leave others to judge for themselves.

My experience has been that more Italian queens get lost in their attempts to meet the drones, than native black or brown queens. Of the superiority of the Italian or Ligurian workers, of their disposition, as well as that of the hybrids, I will speak at some other time. Did it ever occur to you, if the yellow-bearded Italians were natives of our country, and we had been used to looking at them all our lives, and the black were now just discovered and introduced, what praises would be heaped upon the *dub* tails? Campbell uttered a truism when he said—"Tis distance lends enchantment to the view." But do not set me down as against the yellow-jackets. I have been giving them a fair trial for two years—or, rather, an unfair one, for I have tried their strength and weakness, in dividing and subdividing; and when they are reduced to almost a handful, they work with a heroism really commendable.

And right here I wish to say that I think if the Rev. Mr. Briggs, whose article appeared in a former number of the Journal, alludes to queens sent out by Mr. Alley, of Massachusetts, and deems them not reliable by reason

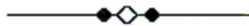
of their low price, he is mistaken. I ordered one from Mr. Alley, and through mistake he sent me two, either one of which, or their workers, will compare favorably with those of anybody. They are not, indeed, as long or as large as your index finger; but I have queens in my yard from various sources, and among them these are the prettiest. Time only will prove the working qualities of the laborers they produce.

WM. P. HENDERSON.

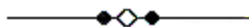
Murfreesboro, Tenn., Aug. 31, 1870.

[NOTE: The Italian queens are, from the brightness of their color, a much more “shining mark” when on the wing, than black queens. Hence, when out on their excursions, they are more liable to be “snapped up” by birds, and doubtless many are thus lost every year. Southern bee-keepers probably suffer more from this circumstance than their northern confreres, as insectivorous birds are more abundant with them.

In some portions of Italy the Ligurian bees were cultivated for centuries, side by side with the common or black bees; yet the difference between them, as regards color or quality, seems to have attracted no attention. But it must be borne in mind that bee-culture fell into decay there, after the fall of the Roman Empire, passing into the hands of a rude and ignorant peasantry. Whereas the superiority of the Ligurians and Cecropians was well known and appreciated in the classic period of the nominal republic. Since the revival of the bee business in Italy (to which it has largely contributed) the Ligurian bee has measurably recovered its pristine favor, and is getting to be preferred everywhere.—ED.]



The yield of honey by various plants and trees depends not only on the character of the season, but on the kind of soil on which they grow.



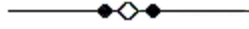
[For the American Bee Journal.]

The Queen Nursery.

As the readers of the AMERICAN BEE JOURNAL are somewhat anxious to hear about the Queen Nursery, invented by Dr. Jewell Davis, of Charleston, Illinois, I will say that it is a perfect success. I have, since the first of June, kept mine running to its full capacity (twelve cages). I have allowed the queens to remain in the cages six or eight days after hatching. I now have his fertilizing attachment, but have not yet tested it. Young, unimpregnated queens can be introduced by Alley's process, to any queenless colony. I will give a fuller report, and how to use it, this fall or winter. I consider it quite an advantage to save all natural queen cells, and hatch them out in the Nursery; and it is no disadvantage certainly to have a supply of young queens on hand, at so small an expense, to give to a natural or artificial swarm, at swarming time, even if they are not fertilized. When you can draw on your nursery for a queen, at any time at sight, it is quite an advantage; at least I consider it so. It is a positive fact that queens perish in their cells by the thousand, in the natural state, in extremely hot weather. In using the Nursery we can control this matter; for if the weather is extra hot, we place the Nursery in a small colony; and in a large strong one, if the weather is cool. Thus you will see that we have the hatching entirely under our own control, and it is not left to chance. The queen breeder can readily see the advantage of separating all his queen cells as soon as sealed over, and having them perfectly safe. I have kept my Nursery in a medium swarm, where they had a perfect queen breeding at the same time. As I said before, queens can be kept in the Nursery any length of time, with perfect safety. I place a small piece of comb containing honey in the cage, between the tins, then place the cell in the cage in a natural position and fasten it with a pin. A very slight fastening answers, as the bees cannot get at it to gnaw it down.

E. GALLUP.

Orchard, Iowa, July 15, 1870.



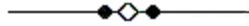
[For the American Bee Journal.]

Paper Hives and Z. C. Fairbanks.

MR. EDITOR:—Don't you think that Mr. Fairbanks seems a little cross as well as sharp. He says I assert in my first article what I contradict in my second on paper hives; and, worst of all, says I am to be numbered with the gentiles, whom Dr. Cox gulled to the tune of heavy sums. I deny the charge, and demand proof; though I will say for the benefit of brother Fairbanks, that I think the Doctor a *little* too smooth for *profit*. But, to explain, we call the paper hive, of whatever form, Dr. Cox's hive; and so should we call all movable frame hives, the Farmer's box with Langstroth frames therein.

CHARLES HASTINGS.

Dowagiac, Mich.



[For the American Bee Journal.]

The Looking-glass Again.

MR. EDITOR:—I have used the looking-glass often for arresting swarms, rarely failing; but *I have always used it in conjunction with the shotgun*. Used thus, it seems to induce in the bees the idea of an approaching storm, and that they ought to be securing a place of safety as quick as possible.

Out of a number of examples, I give the following:

A second swarm proved to be bent on emigrating, for on six consecutive days it left as many different hives. Each time it was brought to anchor by the looking-glass, &c. The last time the bees fell as if shot dead, at the flash and report. And for aught I know and saw, they might have kept trying to this day.

In some rare cases, however, I have failed to bring the swarm to settle.

My bees have swarmed heavily this year, and for a rarity seemed to select the tops of the highest trees to settle on, and then would often leave for the woods after hiving. Query, was there any connection between the two facts?

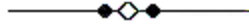
The early season, here, was superior for honey, up to the blooming of the white clover, which was very scarce, and almost devoid of honey. The weather has been hot and dry, and no honey since.

There has been no honey-dew since the war near me; whilst a large piece of woods, three miles off, seemed, two years ago, to be literally flowing with honey-dew, and alive with bees. The tract was three miles wide and five miles long, and alive with bees, throughout its whole extent, every day for several weeks. Did the bees of the country gather there?

Your paper is read with intense interest. Long may it live to contribute to the pleasure and profit of bee-keepers.

J. B. TOWNLEY.

Red Hill Depot, Albemarle Co., Va.



[For the American Bee Journal.]

The Drouth, Bee Pasturage, and Queens.

The honey season has not been good, in this section of country, since the middle of June, in consequence of continued hot and dry weather. Two timely showers served to make a fair crop of corn, but did not much increase the secretion of honey—hence the bees have not gathered more in that period of time than to supply their daily consumption, and keep them brooding. These points I have watched closely. The white clover bloomed nearly two weeks earlier this year, than usual here; and, therefore, by the time the colonies had brooded up to the point of swarming, the chief honey harvest was gone. Hence, but few natural swarms came off, and most of these came near starving to death, and will require doubling up for wintering.

I made a number of artificial swarms, by taking a comb of brood, honey, and bees, from six full hives and putting them together into a new hive—using empty frames to fill the vacancy made in the old hives. The swarms thus made have done well, compared with natural ones, and will be in fair condition for winter.

It continues so dry yet that we cannot look for a large yield of honey, either from buckwheat or other flowers; nor, if we could, can we expect much honey to be stored in boxes, where comb has to be built to receive it, as the nights are becoming too cool for comb-building.

I have seen the bees work incessantly for two or three weeks, this season, upon the plant known as Carpenter's Square, (*SCROPHULARIA NODOSA MARILANDICA*, *Nodose Scrophularia*, *Figwort*,) and also, as usual, on the Purple Polynesia, which appears to yield honey remarkably in hot and dry weather. In this vicinity, also, both the black and the Italian bees have worked on the red clover, during the last weeks of August. But, more than all this, our bees this season seemed compelled to visit the groceries for sugar and other sweets, to supply the lack of honey in the flowers, and have perished by thousands in their demoralized eagerness to obtain them.

From all this we have learned again the *necessity* of cultivating more extensively some crops or plants that will yield honey in the usual barren interval between the failing of the white clover and the Alsike and the coming in of the buckwheat and fall flowers. The linden trees supply this in some localities, but not in ours—being too remote from them. Buckwheat sown about the first of June, will often fill this interval, and that sown a month later will make the fall pasturage. Thus, by a proper disposition of crops, we may, with favorable weather, make a continued honey harvest all the summer months; and, in unfavorable weather, secure at least a partial supply for the same period of time—thereby saving millions of bees from the demoralizing effects of visiting groceries, and the consequent loss of their lives.

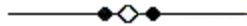
This summer my bees have not been disposed to start as many queen cells as I desired; and, hence, after supplying all my colonies with queens, have not had as many as I wished, to experiment with in the various proposed methods of fertilization in confinement. But I have had enough to show me that under our present knowledge of these processes, none of them are as successful as is desirable for the purposes of the intelligent queen-raiser. I have learned, moreover, that by most of the methods employed the queens and drones become so excited, that, without fostering the disposition for mating (the purpose for which they are confined) they worry themselves to death in a very short time. To remedy this, I have made cages on the same plan of my Queen Nursery cages, but larger every way, with the covered way at one end converted into an *ante-chamber* for the introduction of the drones at the proper season, without disturbing either the workers or the queen in the queen's *parlor*. In this parlor we put two square inches of comb, filled with mature brood, and, over this, three inches square of comb filled with honey for feed; and in the vacant part of it, we suspend a queen cell sealed over. Then, after closing the door, place the cage in a populous stock of bees, for the queen and workers to hatch. Thus, by the time the queen hatches, she will have nearly a hundred workers in the cage with her, and will not become uneasy or excited to get out of the cage. She will thus remain quiet on the comb, until she is old enough to leave it and go in search of the drones. Near this hour the drones can be introduced by the little tin door at the bottom of the ante-chamber, that door closed again and the tin slide carefully removed. The drones and queen are thus let together,

without excitement or disturbance. This cage may be made six inches long, by four inches deep, and one and a half inches wide. Then, by placing the comb in the middle, at the back end of the parlor, with the capped cells facing the wire sides, the bees can emerge from the cells and pass all around the comb.

From various experiments I am led to conclude that the above arrangement will approach nearer to the thing wanted, than any of the plans yet made public. I am, also, further convinced that much attention must be paid to the age of the young queen, and to the state of the weather, in order to secure fertilization in confinement. In fact, we must approach as near as possible to the natural state of the circumstances that govern the mating of queens and drones. I may say, in addition, that it is evident some queens will mate earlier than others, if not hindered by bad weather. The meeting of the queens and drones must not be attended by any circumstances calculated to cause either of them to become alarmed and seek release from confinement; for if thus alarmed or excited, they will worry themselves to death in a few hours, or forget all their natural instinct for mating or fertilization. On the plan above described the queen feels at home where she was hatched, with her hundred associates around her, and under careful management, not liable to become excited. The drones alone are liable to be in any degree alarmed under this method; and I find this is quickly removed by letting them into the presence of a few workers, as in the above case. If done quietly, little excitement need occur.

JEWELL DAVIS.

Charlestown, Ill., Sept. 5, 1870.



[For the American Bee Journal.]

Bee-keeping Advancing.

MR. EDITOR:—We are doing a fine thing in the bee business here this season. We (my brother and I) are creating quite an interest in bee-culture around here, by the use of our Hruschka. The way we sling the honey out is a caution. We have obtained six hundred and twenty-five (625) pounds of extracted honey, and six hundred and fifty (650) pounds of box honey from eight colonies of bees, and have increased them to twenty-two; and all the hives are full of honey now—the result of scientific bee-culture.

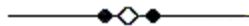
Old foggy bee-keepers begin to open their eyes, and think that bee-keeping is not all mere *luck*. The light begins to shine, and bee-keeping is advancing.

The Italian bees are more and more approved, and taking the place of the black bees; and I am in hopes we shall in a short time have none but Italians around here.

We have tried friend Alley's plan of introducing queens with tobacco smoke, and failed several times, simply because we did not smoke the bees enough. We introduce now successfully with tobacco by smoking them till they are nearly stupefied, and then they will receive the queen without fail. We find the Italians will receive a queen quicker or more readily than the black bees, without any smoking. The Italians are better every way than the blacks. They are as much in advance of the latter as the mowing machine is in advance of the scythe.

D. L. COGGSHALL, JR.

West Groton, N. Y.



[For the American Bee Journal.]

A Visit to Palmer Bros' Apiary, and What I Saw There.

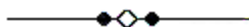
I lately went to visit the apiary of Palmer Bros., at New Boston, in Mercer county. When I came near the house I saw a lot of beehives nicely arranged in rows, north and south, and east and west. They were some eighty in number, I think. The inmates of the house were two very pleasant, clever young men, keeping bachelor's hall. My team was put up and cared for, and we had an interesting talk about bees, beehives, and raising queens.

After dinner the honey-slinger was brought out. It is one of their own getting up, and does well the work it is intended for. A hive was opened, some frames removed, and about twenty pounds of very nice honey slung out in ten minutes.

On returning home and having a good night's sleep, I went into my own apiary next morning with new spirits.

J. BOGART.

Eliza, Ill., Aug. 3, 1870.



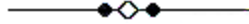
[For the American Bee Journal.]

MR. EDITOR:—You may remember that in the Bee Journal for September, 1869, Mr. George P. Kellogg, of Waukegan, Ill., gave out a very broad challenge to bee-keepers. In the October number, I accepted his challenge; but since that time we have not heard from Mr. Kellogg, through the Journal. Now it is due that he should withdraw his proposition, or meet us at the State Fair, in Michigan, and take an oyster supper, and pay the printer; or cry “*peccavi!*” and I will pay the printer. What say you, brother Kellogg?

We have had an excellent honey season in northern Wisconsin, so far, this summer; with a prospect of its continuing until frost comes. Success to the enterprise, and the Journal.

A. A. HART.

Appleton, Wis., Aug. 6, 1870.



In bee-culture the chief factor is intelligence, and not capital. The former must produce the latter.

THE AMERICAN BEE JOURNAL.

Washington, Oct., 1870.

We have on hand, and unused, numerous favors from correspondents, as most of them having been received too late for this issue. The present arrangements for printing the Journal render it necessary that articles intended for its pages should reach us not later than the 10th of the month, to be in season for the ensuing number.

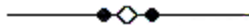
We have received copies of "OLD AND NEW," "EVERY SATURDAY," "GOOD HEALTH," and several other periodicals and publications, which we purposed noticing this month, but are prevented by want of room.

The August number of this Journal contains an article on "*Pure Fertilization Controllable*," translated by the editor from the "Bienenzeitung." It appeared in that sterling and standard periodical, as a communication from the Rev. A. Semlitsch, who is pastor of a congregation and a member of the Ecclesiastical Council at Gratz, in the Austrian province of Stiria. He has been a prominent correspondent of the Bienenzeitung for a quarter of a century, and was previously known as one among the five chief contributors to Vitzthum's "*Monatsblatt für Bienenzucht*," the precursor of the Bienenzeitung. He has always been distinguished for eminent zeal and efficient labor in striving to advance intelligent and scientific bee-culture; and published in 1856, at Gratz, a very excellent practical treatise in aid of the cause. No man in Europe ever questioned his truthfulness, or impeached his honor.

We have copyrighted this Journal, not to prevent or prohibit any of our exchanges from copying articles from its pages, but that those who do copy may see the propriety of giving credit to the AMERICAN BEE JOURNAL, so fully and plainly that there can be no mistake or misapprehension about it. Some have heretofore appropriated such articles bodily and boldly, without giving any credit whatever; some thought they had "*somewhere read*," so and so, &c.; others simply credited "*Ex.*," leaving the whereabouts of the said *Ex.* to be guessed at; others again, extending their liberality a link or two, credit "*Bee Journal*," vaguely and indefinitely. We have borne this hitherto without murmur or complaint, "note or comment," but do not intend to be so forbearing hereafter. If articles are worth copying, their source is worth acknowledging; and those who fail in doing this in future, may expect to have to pay for copyright. We punctiliously give credit ourselves, and may properly ask to receive it.

"Hanc veniam damus petimusque vicissim."

Great waste occurs in feeding meal, in early spring, as a substitute for pollen, and many bees are lost while endeavoring to supply themselves, being chilled by a sudden change of temperature. To prevent this German bee-keepers do the feeding within the hive; and Mr. Kanitz of East Prussia, gives the following as the best mode of doing so: Take fine wheat flour, rye or oat meal, and stir it gradually into lukewarm liquid honey till it forms a pretty stiff paste or mass. In the evening spread a few ounces of this on an empty comb, insert it in a hive, and it will be carried up by the bees in the course of the night. Not more of the paste should be prepared on any occasion, than can be immediately fed. The substitute for pollen thus fed, it is said, greatly promotes brooding.



CORRESPONDENCE OF THE BEE JOURNAL.

RICHMOND, OHIO, August 18, 1870.—I have put off writing till harvest is over, and will now have a short talk with you on different subjects. This summer has been a very pleasant one in this part of the country, with good crops of all kinds except fruit, of which there will be a small yield. We have been favored with plenty of rain and consequently good pasture for stock, and plenty of flowers for the bees which the latter did not fail to enjoy, for they gathered large stores of honey and multiplied more generally than they have done for a number of years.

I have been keeping bees all my life as my father did before me, but never made it a study until about two years ago. Since then I have been trying to put my bees in movable comb hives. These I think every bee-keeper must and will have ere long, as also the Italian bees, which I think are much better than the natives, except that they are inclined to rob the blacks. But I would keep them for their beauty, if they had no other good qualities. I wish some one would give us a general test of their purity as known in Italy. This should be known throughout this country, as nearly every queen breeder has a test of his own. My bees have four bands, counting all; two broad ones next to the middle, and two narrow ones behind those. If this is not enough, then I will go for better and purer ones, as I want the best and none others.

The time of year is coming to think of wintering bees, and I want to build a wooden house large enough to accommodate one hundred hives. I wish some of the knowing ones would give us, through the Journal, proper directions for building such a house.

Now, a few words in conclusion. Inclosed you will find my subscription for the Journal for this year; and please accept my thanks for the valuable instruction I have received from the American Bee Journal, and my best wishes for its success. May its contributors and readers grow wiser and sweeter every year.—J. W. TAYLOR.

BROOKLIN, ONTARIO, August 20.—Bees have done exceedingly well in this Province, this season; better than they have done for several years. Though the loss was fearful last year, it has nearly been made up. This Province is not abundant in forage for bees, and we never expect to realize the figures of *Novice*; yet some have taken from my hive four boxes of virgin honey, eighty (80) pounds; and one hundred and forty-two (142) from the body of the hive, with the Extractor—making *two hundred and twenty-two* (222) pounds from one colony. Another writes me he has taken this season over two thousand (2,000) pounds in boxes, and five hundred (500) pounds with the Extractor.—J. H. THOMAS.

GHENT, OHIO, August 22.—I have read and re-read every number of the Journal, and find it instructive and profitable. My bees wintered well, last winter, in my house as described in Vol. V. page 100, of the Journal. Last winter was with us mild and nice for wintering on summer stands. I have realized two hundred and fifty (250) dollars from thirty hives this season, and have two hundred (200) pounds of honey on hand. It was all box honey. The increase was twenty-five (25) good strong natural swarms. They are all black bees except one, a hybrid queen sent to me last fall, as pure, from an Eastern queen breeder. They are not very sociable. The season was all one could wish for. Bees have done well. The spring opened just right, and continued favorable throughout. Success to you and the readers and columns of the Bee Journal.—T. PIERSON.

ELIZA, ILLS., August 22.—Bees have done well here this season up to this time. I have some in Langstroth hives that have stored one hundred and twenty-five (125) pounds of honey to the hive. I enclose two dollars for the Bee Journal, as I cannot do without it.—J. BOGART.

LEROY, ILLS., August 23.—This is the first year that I have kept bees, and find it a very pleasant business. Bees did not swarm here until August, and then but little. I divided my old stocks in June, all of which, both old and new, are doing finely. I should like to have some older head than mine give me his opinion as to the plan of reducing the number of my stock to one-half this fall, in order to have them stronger and to have plenty of spare comb to commence with in the spring. And, again,—as I am asking favors—I should like to have the plan given on page 109, Vol. IV., B. J., for out-

door wintering republished, for the benefit of new beginners generally as well as myself. The August number came just in time for me to try the new plan of controlling the fertilization of queens. I succeeded in every thing but having the queen mate in the wire case. Will some one else give us his experience? I say three cheers for the American Bee Journal, for I take time to read and re-read every article in it, and find it, together with Mr. Langstroth's valuable book, to be the staff for new beginners to lean upon for apiarian knowledge.—P. YOUNG.

RISING SUN, IND., August 26.—We have a neighbor at Vevay, Mr. W. Faulkoner, who has had great success this season, with his bees. I called on him last week, and had the pleasure of seeing 3,500 lbs. of white clover honey, which with 1,500 lbs. that he has already sold, makes *five thousand* (5,000) *pounds* for this year. He had but forty-eight stands in the spring, so that his hives have averaged over one hundred pounds each. His increase is fifteen stands, making now sixty-three, which is as many as he wants to manage. His hives are a modification of the Langstroth, allowing the use of surplus boxes on the sides of the frames.—N. H. SHAW.

SHREVE, OHIO, August 26.—As I have seen no communication from this place, I have concluded to write and let the readers of the Bee Journal hear of my success in the bee business. I commenced four years ago with the old black bee in the old fashioned way. For a few years I made only slow progress, till of late I have taken more interest in it, and have now increased my stock to seventy-six colonies, all Italians, in good condition.

I was surprised when I read Novice's report of honey this season; but when I came to think over how much I had taken from a few hives with the honey-emptying machine, and as the season was, I think I too could have had a right smart crop, if I had attended to the bees as I should have done in the honey season. As it is, I shall probably not get much over one thousand (1,000) pounds, principally box honey. I will just state, for the benefit of the bee-keeping public, that I have tried a Peabody machine, which works to perfection, and is what every bee-keeper that uses movable frames needs. As far as the different hives are concerned, there is not so much difference as some suppose. I think a plain frame in a simple hive of convenient form is all that any one needs. As far as reliable queen raisers are concerned, I

will just state that I have dealt with a good many, and have found Adam Grimm, of Jefferson, (Wis.,) perfectly reliable and prompt in filling orders. I have got quite a number of queens from him this season by mail, post paid. I inclose a photograph of my apiary, and if any of the readers of the Journal wish one, I will send it on receipt of forty cents, or send one on receiving one for exchange. In conclusion I wish the Journal success, and all its readers good luck and much pleasure in the pursuit of so profitable a business as bee-culture.—G. W. STINEBRING.

EDGEFIELD JUNCTION, TENN., August 29.—This season, thus far, has been the poorest, both for swarming and honey, of any for more than twenty-four years that I have been in this State. We had a drouth in May, followed by frequent and severe cold rains for more than three weeks, by which time our clover harvest for bees was nearly past. As a general thing July and August do not furnish much forage for bees, but we have every prospect for honey this fall. The last two seasons we had a honey harvest from almost the first of April till late in the fall; and on both occasions, late in the fall my hives were so filled with honey that in many of them there were not a hundred empty cells. I removed from one to three frames of honey, placing the remaining frames half an inch or more apart for winter. By doing this, and protecting my hives from the cold winds, I saved them all—one hundred and sixty-four in number last year, and sixty-eight the year before. This season being a poor one, I have not increased stock so much, though I have made fifty-one good colonies. In July I had to feed a few colonies, and found it difficult to keep up my nuclei.—T. B. HAMLIN.

WEST GROTON, N. Y., August 31.—The honey season has been very good here, and scientific bee-culture is progressing. Old fashioned bee-keepers are amazed when they see the large quantity of honey we got from eight colonies of bees—over eight hundred and seventy-five (875) pounds.

I like the American Bee Journal very much. We should not have had near as much honey, if we had not had the Journal to read and study.—D. H. COGGSHALL, JR.

FULTON, ILLS., September 3.—Bees are doing very well here now, though the forepart of the season was not generally favorable on account of the drouth. Buckwheat is not yielding much honey. The second crop of red clover is in

full bloom, and the bees are working on it very busily. This is the first season that I have seen bees do much on red clover, in this section, as the blossom is usually too large; but this year, owing to the drouth the heads are smaller. The different varieties of the golden rod are just coming into bloom, as also the wild aster; and the prospect is that the bees will do well until after we have hard frosts. Light frosts do not affect the aster. If acceptable, I will try to furnish some account of the doings, of the bees in this section, at the close of the season.—R. R. MURPHY.

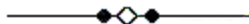
GENOA, ILLS., September 9.—Please excuse my being thus dilatory in not making an earlier remittance for the Journal. This little amount I could have turned to very good account in other directions; yet, as I am circumstanced, I think that one volume of the American Bee Journal is worth three or four times as much to me as the same sum laid out in any other way at home. For had it not been for the Journal, I should long since have been as many of my neighbors are—“*one that USED to keep bees.*” I am aware that my location is not naturally favorable for bee-keeping, as we sometimes have two or three seasons in succession that are hard on the bee business; yet I am not inclined to give it up so. In 1868, I put twenty swarms into my kitchen cellar. Most of them had not one pound of honey on the first of January; but I made up my mind to try the winter feeding to my full satisfaction. I took off caps, cut a hole two inches by five through the honey board, which was half an inch thick; fastened cotton cloth upon the under side, which made a box large enough to hold all the food I wished to put in at a time. The food was syrup of good refined sugar. I took care that they were all ventilated according to the size of the stock; and as the temperature would change in a measure with that outside, I would regulate ventilation accordingly; and by constant attention they come out in the spring with the loss of only two swarms, besides two that became queenless. No more bees died than usual in wintering; and although the season last year was wet and cold, they managed to procure sufficient to carry them through the winter in tolerably good condition. But this spring and summer the drouth seemed to threaten them with starvation. We had no rain from the last of March till the first of July, with the exception of two slight showers that did not, either of them, wet the ground more than an inch deep. Notwithstanding, with the white clover, which put out some small blossoms and in moist places where not pastured, continued fresh, and some wild flowers, the bees kept along

till the rains came in July. Then the clover and other blossoms came out quite fresh; so for a few weeks the bees gained a little and afforded some surplus honey. Now the buckwheat is in full bloom, and the bees seem to be taking time by the foretop, by improving each hour, shine or no shine. The hybrid bees, as well as the pure-blooded, appear to be exerting themselves to vindicate the superior merits of their ancestors; and although it may seem cruel, I stand ready, with open and greedy hands to receive their hard-earned stores, and furnish them with store-room to enable them to continue on another willing task. My eighteen acres of Alsike and two of melilot clover are entirely killed by the drouth. For three years I have not only had to contend with adverse seasons, but have been a target for friends and neighbors to pop their jokes at, for my persistence in such unprofitable business. But I had made up my mind to fight it out on this line; and by the assistance of the American Bee Journal, with its able and generous contributors, am confident that eventually I will come out all right. Though the season has been a hard one, I have now taken out honey enough to pay for all the sugar I have used and for the four volumes of the Journal, and have added one-third to the number of my stocks this season—while many old fogies of my acquaintance, who laugh at the idea of using patent hives or paying the trifling sum for the Journal, have lost some nearly all, and others quite all of their bees.—A. STILES.

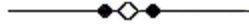
SPARTA CENTER, MICH., September 7.—I cannot think of getting along without the Journal. I supposed that I was doing extremely well in the bee business, until I read Novice's reports, which are surprising. I have kept bees four years, commencing with nine colonies in box hives. At the end of the first season, I had fourteen, all told. I buried them according to the plan recommended in Langstroth's "Hive and Honey Bee," and lost two. The second summer I had fourteen new swarms, making my stock twenty-six in the fall; but, as the season was a poor one, I had no surplus honey. I buried them in clumps, as before, and in the spring found three were *non est*. This was the spring of 1869. During the ensuing summer, I had twenty-four new swarms and nine hundred (900) pounds of surplus honey, and began to know something of the habits, &c., of bees. In the fall of 1869, I built a bee house for wintering, 10 feet by 20, outside measure, 8 feet by 18 inside. The walls were made by using two rows of studding, boarded up outside and inside of each row, leaving an air space between the walls, and filling

between the studding with saw dust. This spring I had forty-six good stocks, and have obtained 2,194 pounds of No. 1 honey. I have now one hundred and ten (110) colonies, all but three or four in good condition for wintering. I have no Italian bees, as I wished to learn to manage and handle the blacks, before trying any that might require more skill. I use Langstroth's "shallow things." All except five of my swarms are in frame hives, and every comb is straight with not over sixteen square inches of drone comb to a hive. Sixty-nine of my queens are of the present season. All my new colonies were made artificially, except six. I made them by starting nuclei, and building up by taking comb, honey, and brood from strong stocks. I fed each colony a little syrup every alternate day from April 1 to June 1. Nearly all the surplus honey of this year is made from or *gathered* from white clover blossoms. Last year it was from linden or basswood.—I should like to know if Novice or others using the melextractor, have had any trouble with the honey fermenting after being canned. I have had several cans spoil. It assumed a reddish hue and became watery in appearance. I should like to know how to avoid losing any in future.—A. B. CHENEY.

WINCHESTER, VA., September 10.—This has been a good season for honey, but few swarms. I started in the spring with sixty-four colonies and have had twenty-one swarms. They will make a fine lot of honey. I use the Langstroth hive. Some of my neighbors that have ten or twelve old-fashioned box hives, think the Langstroth hive costs too much, but come to me every fall to buy honey. I have seven colonies of Italian bees. I think they are superior to the black bee, both for swarming and making honey. I obtained my queens of Mr. Henry Alley. I think he deserves great credit for sending pure queens and acting honorably with his patrons. My bees are not making any honey now, as there was no buckwheat sown in this part of the country. The most that we have to depend on in this country is white clover and blue thistle. We sowed one pound of Alsike clover seed in April, 1869, and mowed it for seed July 25, 1870. I thought it a humbug, but am agreeably disappointed. My bees worked on it from early morn till late at night. The farmers are much pleased with it, both for hay and pasture.—B. F. MONTGOMERY.



It cannot be too deeply impressed on the mind of the bee-keeper, that a small colony should be confined to a small space, if we wish the bees to work with the greatest energy, and offer the stoutest resistance to their numerous enemies.—*Langstroth*.



[For The American Bee Journal.]

Two Queens in One Hive.

When removing some honey boxes on the 25th of July last, I found that a large strong stock had two queens. I see in Vol. V., No. 8, of the Journal, that Mr. E. M. Johnson discovered two queens in one of his hives in January. Before movable comb hives were used to any great extent, such a thing was considered impossible; but we hear of such cases frequently, now that we have easy access to the interior of our hives.

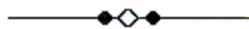
After removing the boxes, I placed them in my cellar, to have the bees go back to their hives; which they all did, except those in one box, which I found contained the queen that I had saved about a fortnight before, a few days after they had swarmed. In removing a frame of brood to give to a weak stock, when brushing off the bees in front of the hive, I saw there was a fine looking queen with them. She went into the hive and was received by the bees. Now, why was this queen in a box containing sealed honey? I should judge both queens were fertile. The bees had killed off their drones a number of days before, so that they did not think of swarming.

Now can we say positively that two queens are not tolerated in one hive? Is it not possible that the workers cluster around them, and keep them apart?

The next day, I returned the queen, after smoking both queen and bees. She was well received, and was all right the next time I opened the hive; and for all I know, they have two queens still. If other bee-keepers have such cases, I should be pleased to hear from them through the Journal.

A. GREEN.

Amesbury, Mass., Aug. 15, 1870.



[For The American Bee Journal.]

Bee Houses.

MR. EDITOR:—It is now admitted that bee houses are requisite for bee-keepers in this climate.

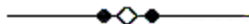
I have recently seen that “concrete buildings” are “cheap and substantial. For dwellings, all hollow walls and lathing are dispensed with,” and they are “found to be as dry as wooden houses.” It is also said—“The heat would be so long retained in the walls, that the saving in fuel would be no inconsiderable item.”

It appears to me that this is just what is wanted in those localities where the material can easily be had.

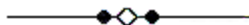
Will some of your correspondents, acquainted with the subject, give an opinion as to their adaptability, and mode of construction?

TYRO.

Ontario, Canada.



The blossoms of onions abound in honey, the odor of which, when first gathered, is very offensive; but before it is sealed over, this disappears.—
LANGSTROTH.



[For the American Bee Journal.]

Bees in Hancock County, Indiana.

MR. EDITOR:—Having been raised in the mountains of Virginia, I had not much chance for schooling and do not expect to write anything smart; but in my blundering manner will try to tell you how I am getting along in the bee business.

In the fall of 1868, I had twelve stands of black bees in log and box hives. All seemed to be in nice order and doing well. But they became subject to dysentery, flux, or whatever you may please to call it. The disease did its work, and next spring I had one colony left, with not over a quart of bees. But 1869 was a good season for bees. My one colony cast five swarms, and the first swarm cast one also—making seven in all. All wintered well on their summer stands.

This spring I bought Langstroth hives, and on the 27th of May got a man that understood the business to come and help me transfer and divide them. We put them in fourteen hives, and all are doing well. We took away the black queens and gave them Italian queens—one of which died or was killed before commencing to lay, for which my man sent me another in her place. Another either died or was killed, nine days after she was introduced, but left plenty of young brood; and they have not less than fifteen queen cells capped and nearly ready to hatch. Query, would it be better to divide them as they are very strong, and then have their queens fertilized by black drones, as I have no Italian drones yet? Or should I let them alone, and let them swarm or kill off all their queens but one, as they see fit?

I intend to divide all my bees as soon as Italian drones are plenty. Mine are the only Italian bees in this settlement, and the woods are full of black bees. I shall be troubled with hybrids, but intend to keep on in the good work until I have them all pure Italians.

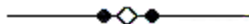
Our country is almost covered—that is, pastures and meadow—with white clover. Even the lanes and highways are white with its bloom, and bees

have a good time gathering honey.

I am well pleased with the JOURNAL, and add the names of some bee-keepers, who perhaps do not yet take it. I think you would do well to send them specimen numbers.

JONATHAN SMITH.

Willow Branch, Ind.



BEES ALOFT.—About two years ago, a swarm of bees was discovered in the steeple of the Congregational Church in Gilsum, N. H., where they have since remained. As a result, fifty-six pounds of honey were recently obtained from the sacred edifice.—*Boston Journal*.



Light colonies, deficient in honey, should be fed in the latter part of September or early in October. If feeding is begun early, in seasons where late forage is abundant, there will be a great waste of honey.—*Langstroth*.

Transcriber's Note:

Obvious printer errors corrected silently.

Inconsistent spelling and hyphenation are as in the original.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
BEE JOURNAL, VOL. VI, NO. 4, OCTOBER 1870 ***

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